7 Powers

The Foundations of Business Strategy

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Foreword by Reed Hastings

Audiobook Supplement

ECHO POINT BOOKS & MEDIA, LLC

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Value

So far in this chapter I have separately defined "Strategy" and "strategy." The first tied back to value, the second to Power.

As an Economist it is my habit to use some light math to bring clarity to such definitions. In what follows, I establish the link between "Strategy" and "strategy" by mapping value to my definition of strategy.

For the purposes of this book, "value" refers to absolute fundamental shareholder value[5]—the ongoing enterprise value shareholders attribute to the strategically separate business of an individual firm. The best proxy for this is the net present value (NPV) of expected future free cash flow (FCF) of that activity.[6]

$$\mathbf{NPV = \Sigma(CF_i/[1+d]^i)}$$

Where:

CF_i = expected free cash flow in period i

d = discount rate

A mathematically equivalent[7] but more felicitous formula for the NPV of free cash flow is:

$$\mathbf{NPV = M_0\, \mathit{g}\, \bar{\mathit{s}}\, \bar{\mathit{m}}}$$

Where:

M_0 = current market size

g = discounted market growth factor

$\bar{s}$ = long-term market share

$\bar{m}$ = long-term differential margin (net profit margin in excess of that needed to cover the cost of capital)

So: $$\boxed{\mathbf{Value = M_0\, \mathit{g}\, \bar{\mathit{s}}\, \bar{\mathit{m}}}}$$

I refer to this as the **Fundamental Equation of Strategy.** Recall my definition of strategy:

strategy: *a route to continuing Power in significant markets*

Appendix to Introduction: Derivation of the Fundamental Equation of Strategy

Definitions

$\pi_i \equiv Profits\ in\ period\ i\ (after\ taxes, before\ interest)$

$I_i \equiv Net\ investment\ in\ period\ i$

$= \Delta\, Working\ Capital + Gross\ fixed\ investment - Depreciation$

$K_i \equiv Capital\ at\ end\ of\ period\ i$

$K_0 \equiv Initial\ capital$

$P \equiv Terminal\ Sale\ Price$

$c \equiv Cost\ of\ capital$

$r \equiv Rate\ of\ return$

$\gamma \equiv Differential\ return = r - c$

$\eta \equiv Top\ line\ growth$

$CF_i \equiv Cash\ flow\ in\ period\ i = \pi_i - I_i$

Net Present Value

$$NPV = -K_0 + \sum_{i=1}^{i=n} \frac{\pi_i - I_i}{(1+c)^i} + \frac{P}{(1+c)^n}$$

$$= -K_0 + \sum_{i=1}^{i=n} \frac{\pi_i - (K_i - K_{i-1})}{(1+c)^i} + \frac{P}{(1+c)^n}$$

$$= -\text{Initial investment} + \text{Discounted cash flows} + \text{Discounted terminal value}$$

$$= \sum_{i=1}^{i=n} \frac{\pi_i}{(1+c)^i} - K_0 + \sum_{i=1}^{i=n} \frac{-(K_i - K_{i-1})}{(1+c)^i} + \frac{P}{(1+c)^n}$$

$$= \sum_{i=1}^{i=n} \frac{\pi_i}{(1+c)^i} - \left[K_0 + \sum_{i=1}^{i=n} \frac{K_i - K_{i-1}}{(1+c)^i}\right] + \frac{P}{(1+c)^n}$$

Simplifying the Middle Term

$$K_0 + \sum_{i=1}^{i=n} \frac{(K_i - K_{i-1})}{(1+c)^i} = K_0 + \frac{K_1 - K_0}{(1+c)} + \frac{K_2 - K_1}{(1+c)^2} + \cdots + \frac{K_n - K_{n+1}}{(1+c)^n}$$

$$= K_0 - \frac{K_0}{(1+c)} + \frac{K_1}{(1+c)} - \frac{K_1}{(1+c)^2} + \frac{K_2}{(1+c)^2} - \cdots + \frac{K_{n-1}}{(1+c)^{n-1}} - \frac{K_{n-1}}{(1+c)^n} + \frac{K_n}{(1+c)^n}$$

$$= K_0\left(1 - \frac{1}{1+c}\right) + K_1\left(\frac{1}{1+c} - \frac{1}{(1+c)^2}\right) + \cdots + K_{n-1}\left(\frac{1}{(1+c)^{n-1}} - \frac{1}{(1+c)^n}\right) + \frac{K_n}{(1+c)^n}$$

$$= \frac{K_0}{(1+c)^0}\left(1 - \frac{1}{1+c}\right) + \frac{K_1}{(1+c)^1}\left(1 - \frac{1}{1+c}\right) + \cdots + \frac{K_{n-1}}{(1+c)^{n-1}}\left(1 - \frac{1}{1+c}\right) + \frac{K_n}{(1+c)^n}$$

$$= \frac{K_0}{(1+c)^0}\left(\frac{1}{1+c}\right) + \frac{K_1}{(1+c)^1}\left(\frac{1}{1+c}\right) + \cdots + \frac{K_{n-1}}{(1+c)^{n-1}}\left(\frac{1}{1+c}\right) + \frac{K_n}{(1+c)^n}$$

$$= K_0 \frac{c}{1+c} + K_1 \frac{c}{(1+c)^2} + \cdots + K_{n-1} \frac{c}{(1+c)^n} + \frac{K_n}{(1+c)^n}$$

$$= \sum_{i=1}^{i=n} K_{i-1} \frac{c}{(1+c)^i} + \frac{K_n}{(1+c)^n}$$

This is the simplified middle term. Substituting this back:

$$\Rightarrow NPV = \sum_{i=1}^{i=n} \frac{\pi_i}{(1+c)^i} - \left[\sum_{i=1}^{i=n} K_{i-1} \frac{c}{(1+c)^i} + \frac{K_n}{(1+c)^n}\right] + \frac{P}{(1+c)^n}$$

$$= \sum_{i=1}^{i=n} K_{i-1} \frac{\pi_i - cK_{i-1}}{(1+c)^i} + \frac{P - K_n}{(1+c)^n}$$

Assume the business has a finite life = L. At t = L, P = 0. So ∃ an n* < L ∋ $\left|\frac{P-K_n}{(1+c)^n}\right| < \epsilon$ with an ϵ that is not material to NPV. Thus at n* we can ignore the second term $\frac{P-K_n}{(1+c)^n}$. Thus

$$NPV = \sum_{i=1}^{i=n*} K_{i-1} \frac{\pi_i - cK_{i-1}}{(1+c)^i}$$

$$= \sum_{i=1}^{i=n*} \frac{rK_{i-1} - cK_{i-1}}{(1+c)^i}$$

$$= \sum_{i=1}^{i=n*} \frac{K_{i-1}(r-c)}{(1+c)^i}$$

$$= \sum_{i=1}^{i=n*} \frac{K_{i-1}\gamma}{(1+c)^i}$$

$$= \sum_{i=1}^{i=n*} \frac{K_0(1+\eta)^{i+1}}{(1+c)^i}\gamma$$

$$= K_0 \sum_{i=1}^{i=n*} \frac{(1+\eta)^{i+1}}{(1+c)^i}\gamma$$

$$\Rightarrow \text{NPV} = K_0 g\gamma \qquad where\ g \equiv the\ discounted\ growth\ factor = \sum_{i=1}^{i=n*} \frac{(1+\eta)^{i+1}}{(1+c)^i}$$

First-period profits in excess of their capital cost $= K_0\gamma$

Alternatively,

$K_0\gamma = M_0\, \bar{s}\, \bar{m}$ (both are expressions of first-period profits in excess of the cost of capital)

Where:

$M_0 \equiv initial\ market\ size$

$\bar{s} \equiv average\ market\ share$

$\bar{m} \equiv average\ profit\ margin\ above\ that\ needed\ to\ return\ the\ cost\ of\ capital$

Therefore, we can write:

$NPV = M_0\, g\, \bar{s}\, \bar{m}$ Which can be interpreted:

$NPV = Market\ Scale \cdot Power$

The derivation of the Fundamental Equation of Strategy thus used these simplifying assumptions:

1. n = n*
2. market growth is constant over n*
3. market share is constant over n*
4. differential returns are constant over n*
5. the business has a finite life

If one were to compare the value derived to actual market cap, in addition to altering these assumptions, one would have to add back the initial capital, account for price levels in the overall market, and add back excess assets on the balance sheet (accumulated cash for example).

Figure 1.1: Blockbuster and Netflix Revenue[15]

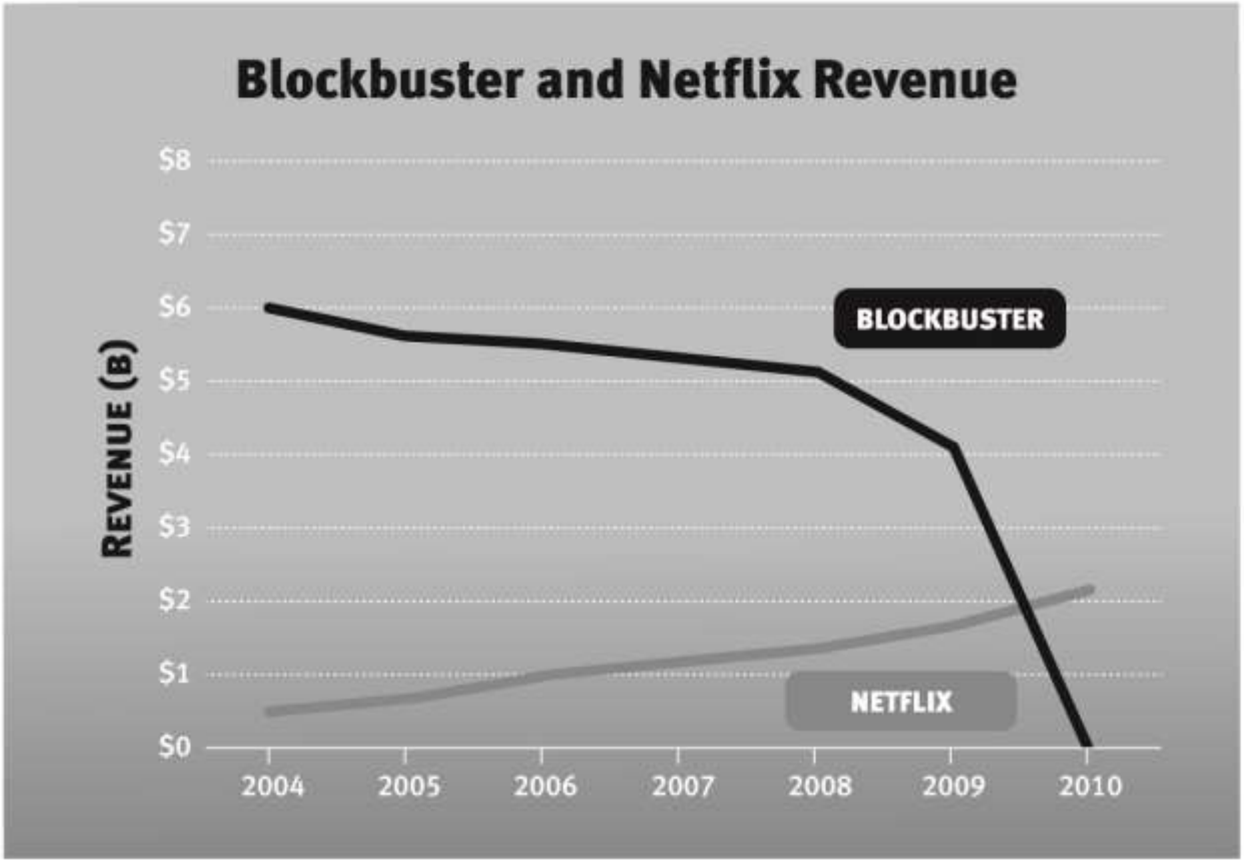


Figure 1.2

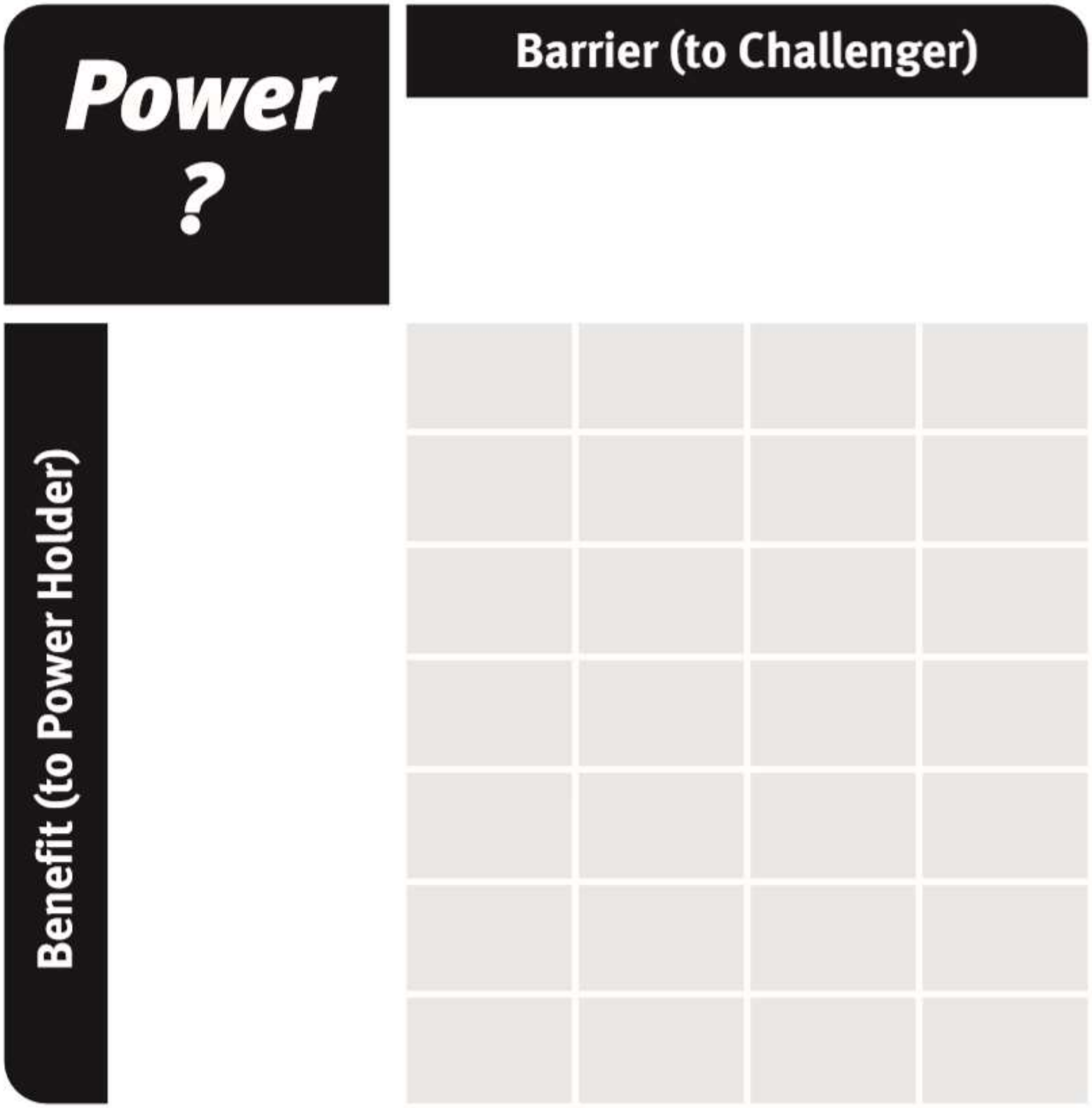


Figure 1.3

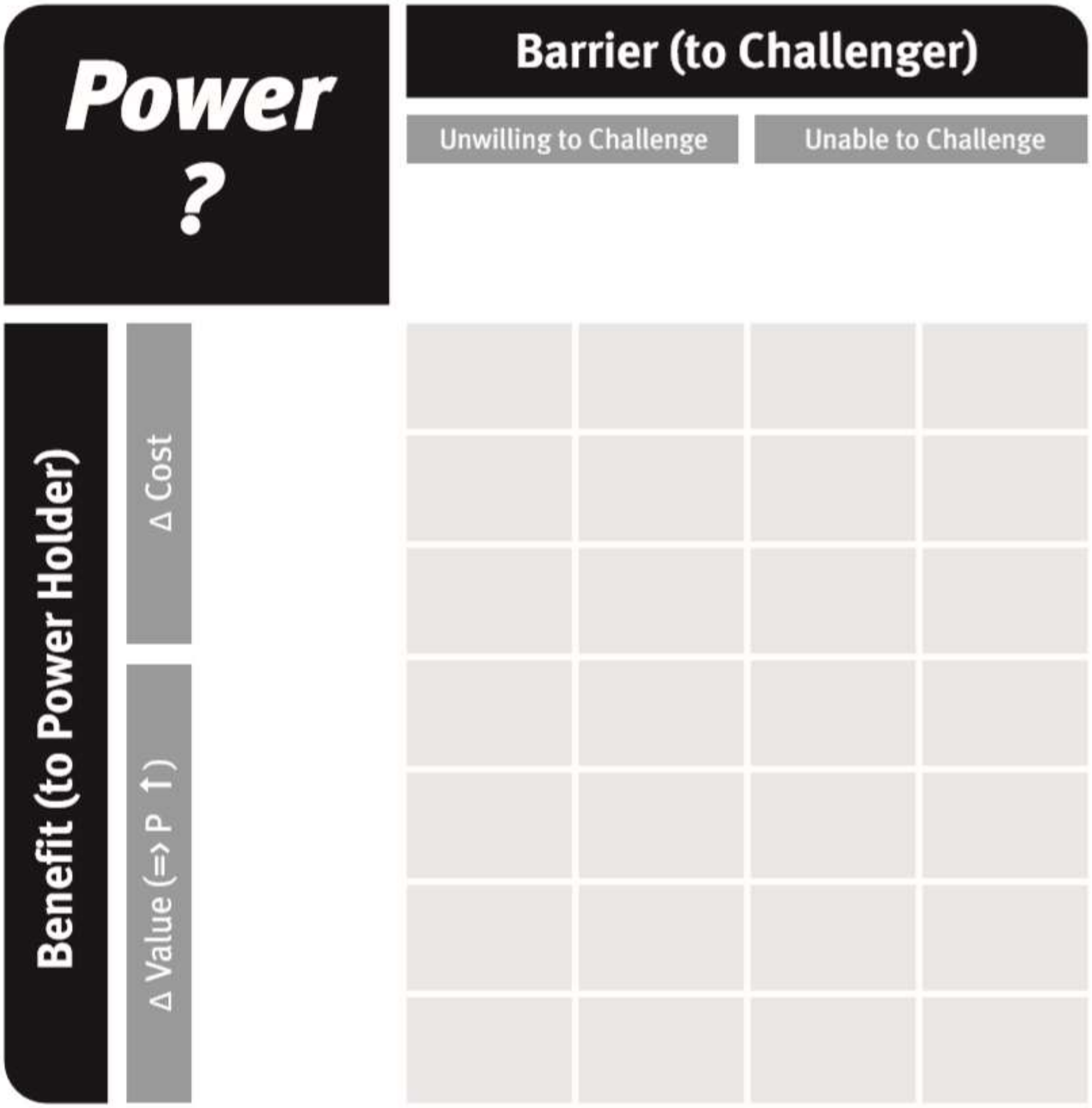

Power
?
Barrier (to Challenger)
Unwilling to Challenge
Unable to Challenge
Benefit (to Power Holder)
Δ Cost
Δ Value (=› P ↑)

Figure 1.4: Scale Economies in the 7 Powers

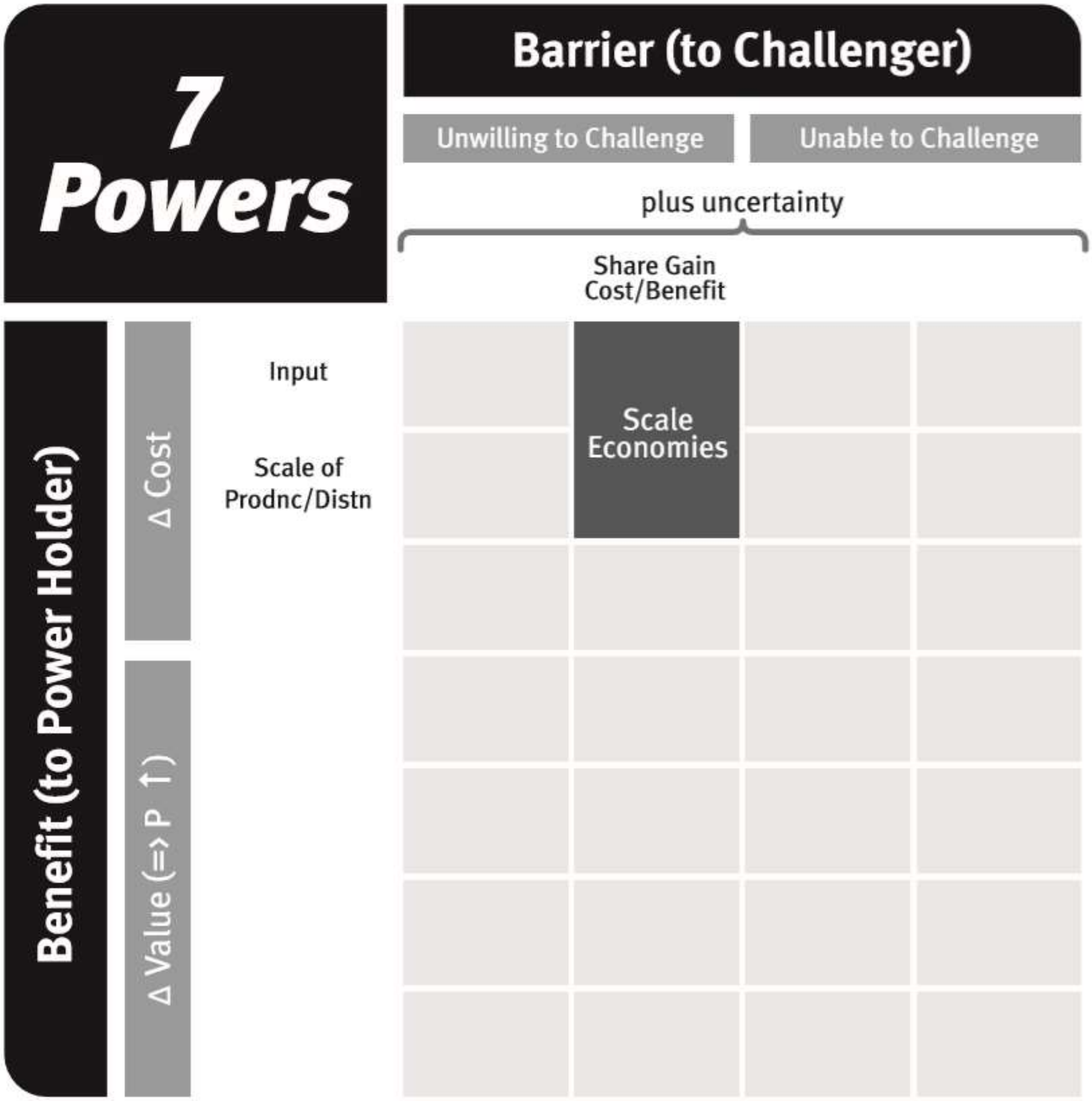


Figure 1.5: Netflix Stock Price vs. S&P 500 TR (8/2010 = 100%)[18]

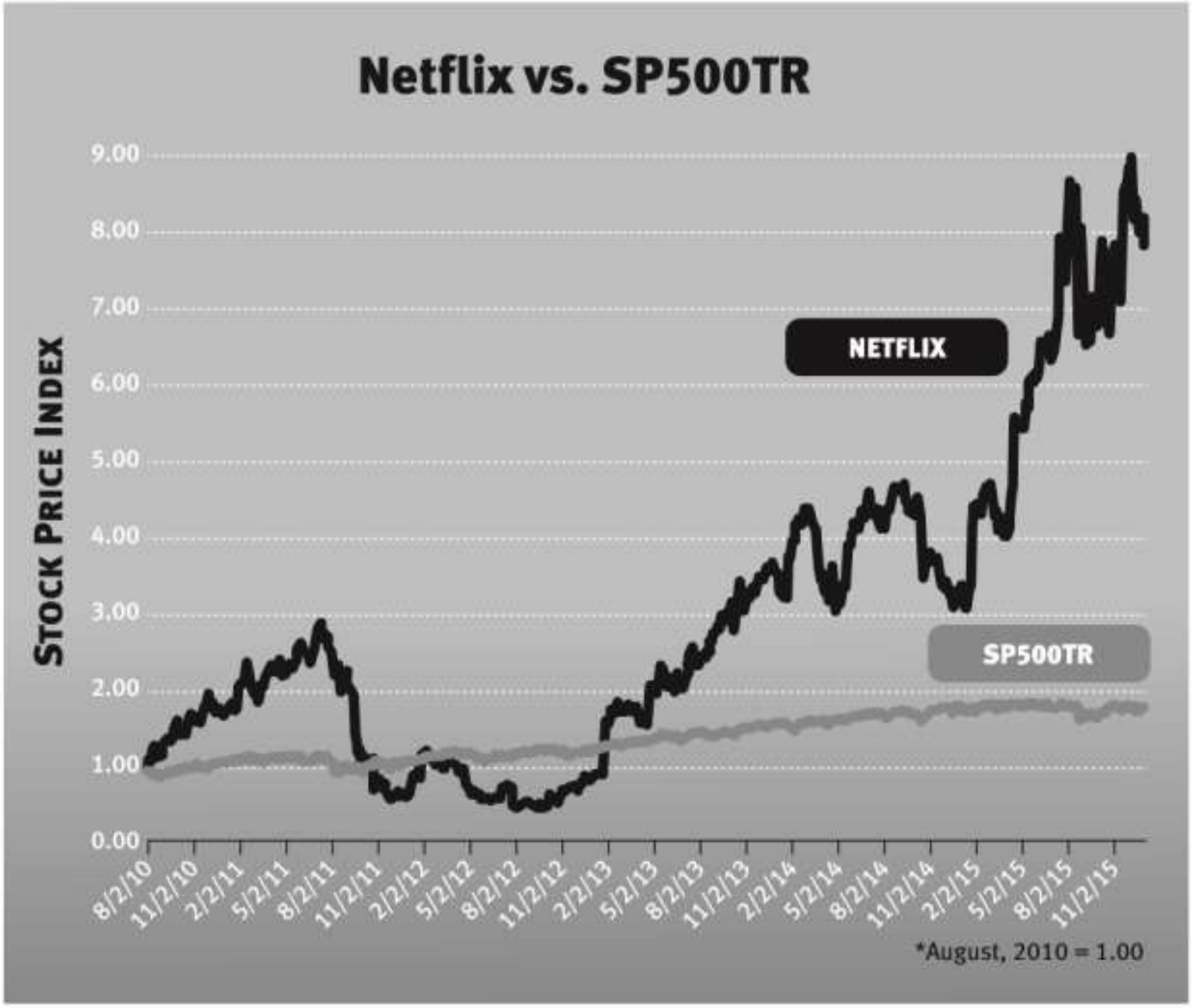


Figure 1.6: Power Intensity Determinants

	Industry Economics	**Competitive Position**
Scale Economies	Scale economy intensity	Relative scale

Appendix 1.1: Derivation of Surplus Leader Margin for Scale Economies

To calibrate the intensity of Power, I ask the question "What governs profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?"

This appendix explores Scale Economies from a fixed cost. There are sources of Scale Economies other than a fixed cost but this is a common one.

Total cost = c Q + C

Where c ≡ variable cost per unit

Q ≡ units produced

C ≡ fixed cost (during each production period as opposed to start-up)

∴ Profits (π) = (P – c) Q – C

Where P ≡ price faced by all sellers

There are two businesses: S, the strong company, and W, the weak company

As an indication of leader leverage, assess:

Surplus Leader Margin: What governs S's margins if P is set $\ni {}_W\pi = 0$?

$${}_W\pi = 0 \Rightarrow \quad 0 = (P - c)\,{}_WQ - C$$

$$\text{or} \quad P = c + C/{}_WQ$$

$${}_S\pi \quad = (P - c)\,{}_SQ - C$$

Substituting in formula for P

$$= ([c + C/{}_WQ] - c)\,{}_SQ - C$$

$$= [C/{}_WQ]\,{}_SQ - C$$

$$= C\,({}_SQ - {}_WQ)/{}_WQ$$

or

$$\text{S' margin} \equiv {}_S\pi/{}_S\text{Revenue} = {}_S\pi/(P\,{}_SQ) = [C/(P\,{}_SQ)][\,({}_SQ - {}_WQ)/\,{}_WQ\,]$$

$$\boxed{\textbf{Surplus Leader Margin} = [C/(P\,{}_SQ)]\,[{}_SQ/{}_WQ - 1]}$$

$[{}_SQ/\,{}_WQ - 1]$: Competitive Position—relative market share beyond parity

$[C/(P\,{}_SQ)]$: Industry Economics—the relative importance of the fixed cost

Figure 2.1: BranchOut Monthly Active Users (millions, January 1, 2012–June 23, 2012)[23]

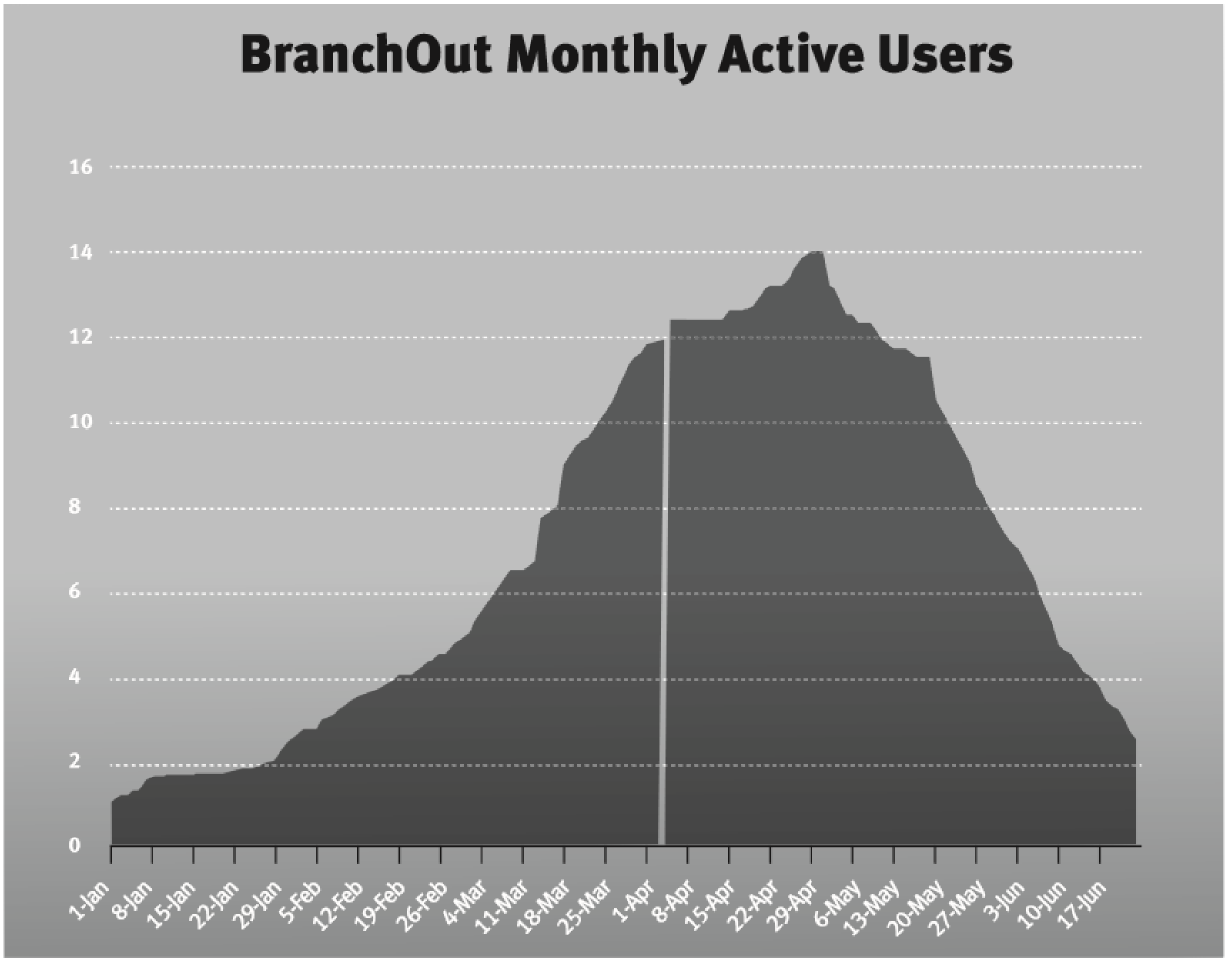


Figure 2.2: Network Economies in the 7 Powers

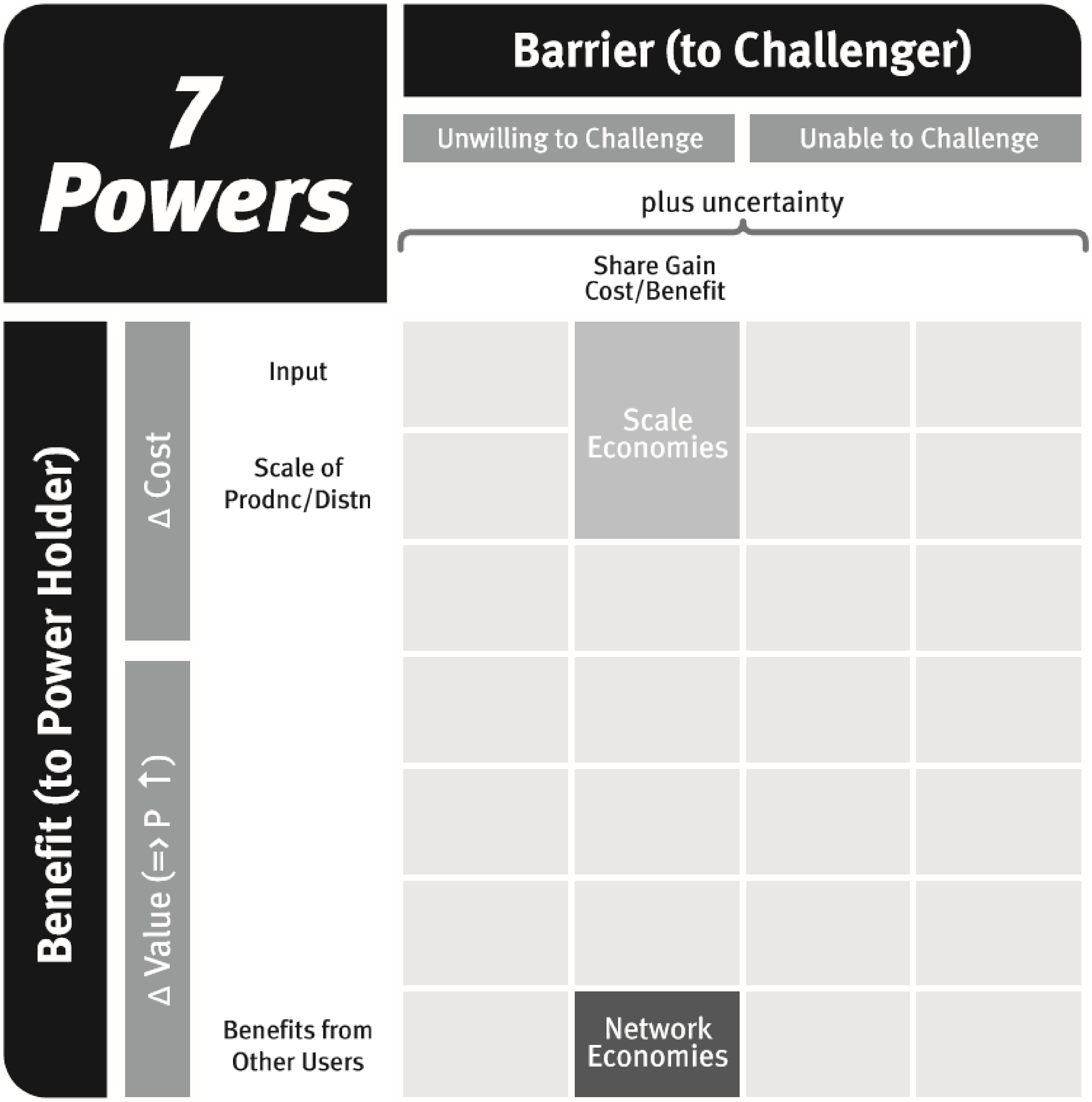


Figure 2.3: Power Intensity Determinants

	Industry Economics	***Competitive Position***
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Intensity of network effect	Absolute difference in installed base

Appendix 2.1: Derivation of Surplus Leader Margin for Network Economies

To calibrate the intensity of Power, I ask the question "What governs profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?"

The total network size (#users) $\equiv N = {}_SN + {}_WN$

Where S is the strong company and W the weak company

Suppose for simplicity the network effects are homogeneous; then S is able to charge a price premium:

${}_SP - {}_WP = \delta\,[{}_SN - {}_WN]$ where $\delta \equiv$ the marginal benefit to all users from one joiner

There are no scale economies so

Profit in a time period $\equiv \pi = [P - c]\,Q$

with $P \equiv$ price

$c \equiv$ variable cost per unit

$Q \equiv$ units produced per time period

As an indication of leverage, assess:

What governs S's margins if P is set $\ni {}_W\pi = 0$?

${}_W\pi = 0 \Rightarrow \quad 0 = (P - c)\,{}_WQ \quad \Rightarrow {}_WP = c$

S can charge a premium so, ${}_SP = \delta\,[{}_SN - {}_WN] + c$

$\therefore \quad {}_S\pi = [(\delta\,[{}_SN - {}_WN] + c) - c]\,{}_SQ$

${}_S\pi = [\delta\,({}_SN - {}_WN])\,{}_SQ$

Surplus Leader Margin $\equiv {}_S\text{Margin} = [\delta\,({}_SN - {}_WN])/[(\delta\,[{}_SN - {}_WN] + c)]$

${}_S\text{Margin} = \delta\,[{}_SN - {}_WN]/[(\delta\,[{}_SN - {}_WN] + 1]]$

$$\boxed{\textbf{Surplus Leader Margin} = 1 - 1/[(\delta/c)\,({}_SN - {}_WN) + 1]}$$

Competitive Position: $[{}_SN - {}_WN]$—Absolute difference in installed base

Industry Economics: δ/c—The value increases with each additional user per dollar variable cost

If ${}_SN = {}_WN$, then SLM = 0; as ${}_SN >> {}_WN$, SLM -> 100% with $\delta>0$

Some comments on Network Economies:

- There can be positive network effects but no potential for Power.
  - The network effect δ needs to be large enough relative to the potential installed base and the cost structure for there even to be one profitable player as this fulfills the Benefit condition. If homogeneous network effects are the only value source, then if $N \delta < c$, a firm cannot reach profitability.
  - This is the problem I see often in Silicon Valley. If one supposes Network Economies then the strategy imperative is to scale much faster than anyone else—if another firm gets to the tipping point before you, then the game is over.
  - However, *ex ante* it is often very difficult to have much assurance in sizing potential N and δ. So you are left with a situation that sometimes requires significant up front capital but an uncertain ability to monetize. This for example has plagued Twitter. Usually management gets the blame but we are back to Buffett's observation: "When a manager with a reputation for brilliance tackles a business with a reputation for bad economics, the reputation of the business remains intact."[25]
- Network effects can be very complex. As indicated earlier, however, there are many excellent treatments so I have been brief. A common twist I have not covered are indirect network effects (also called demand side network effects).
  - If a business has important complements and these complements are somehow exclusive to each offering, then a leader will attract more and/or better complements.
  - As a result the entire value proposition to a customer is improved (increasing SLM for example).
  - An example of this would be smartphone apps. Another smartphone OS would be hard to offer at this point because it would start out with

a dearth of apps. This would make it very unattractive. Developers of apps would of course not be incented to spend their scarce resources since the market would be small.

- Note that in this case the contribution of additional complements is not linear.

Figure 3.1: Share of Actively Managed US Equity Funds that Beat the US Equity Market[26]

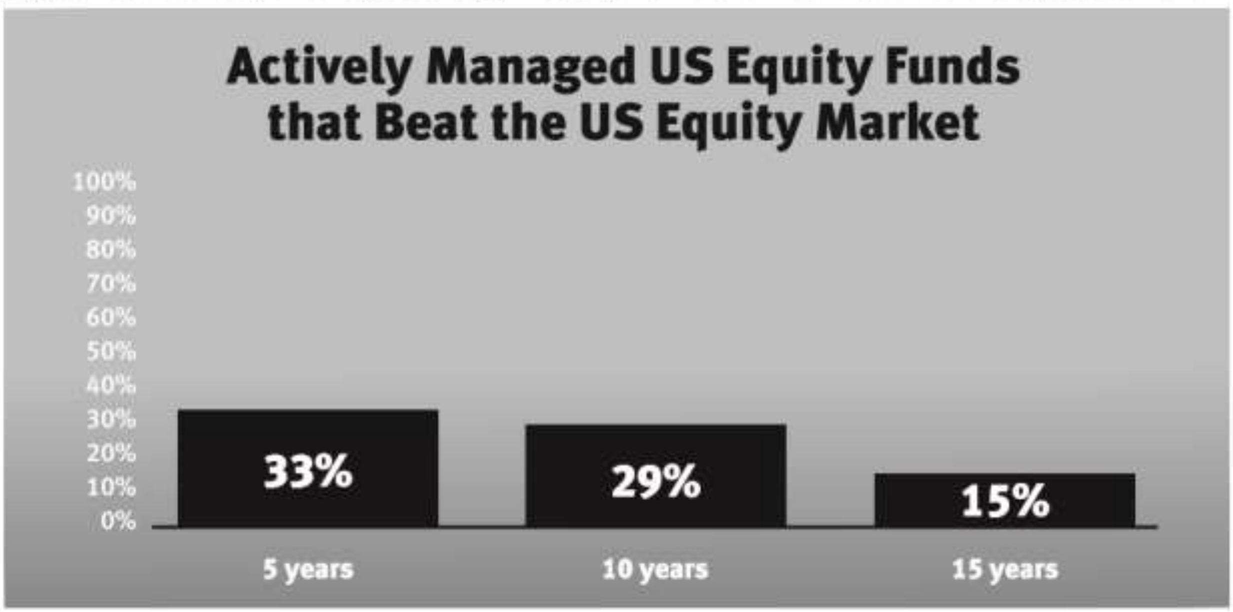


Figure 3.2: Vanguard Assets Under Management (1975–2015)[27]:

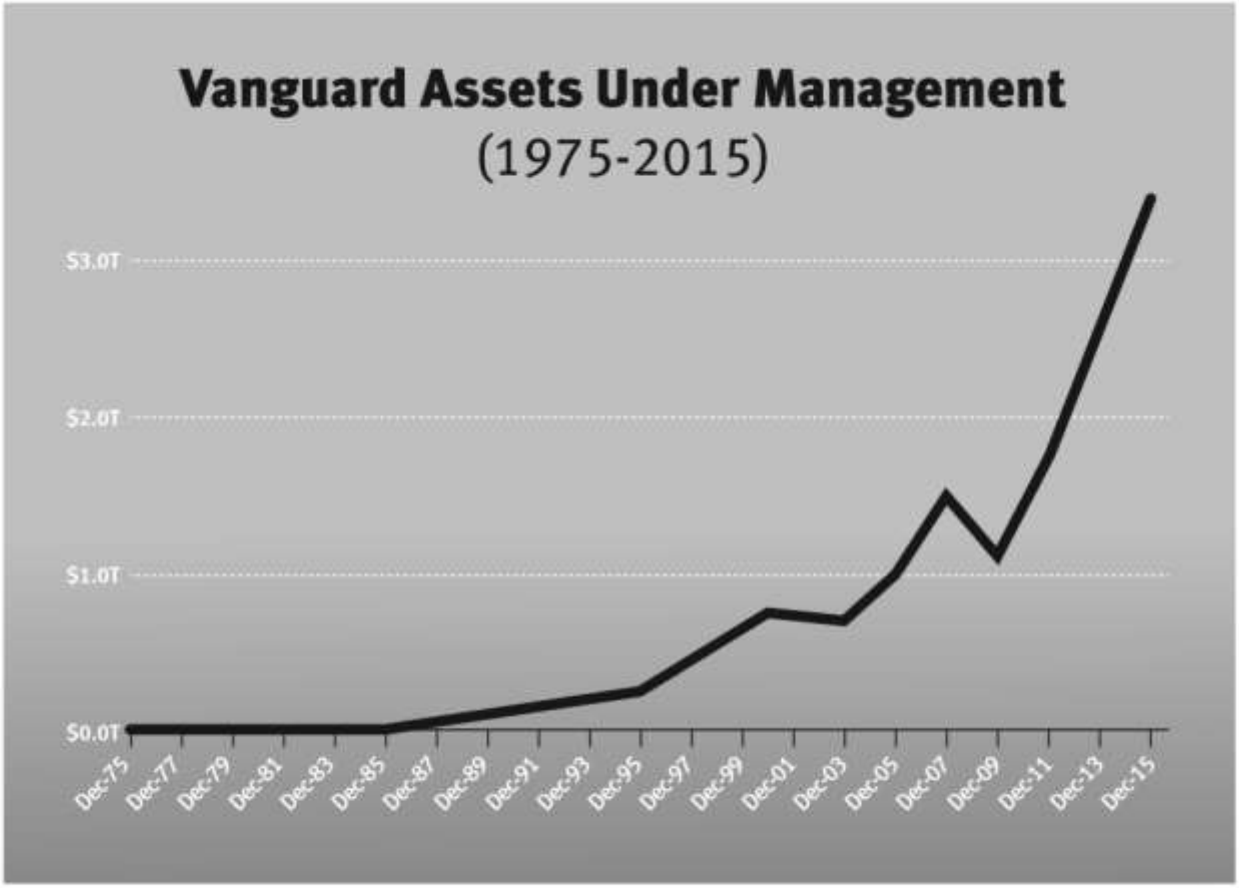


Figure 3.3: Cumulative Investment Flows by Fund Type[28]

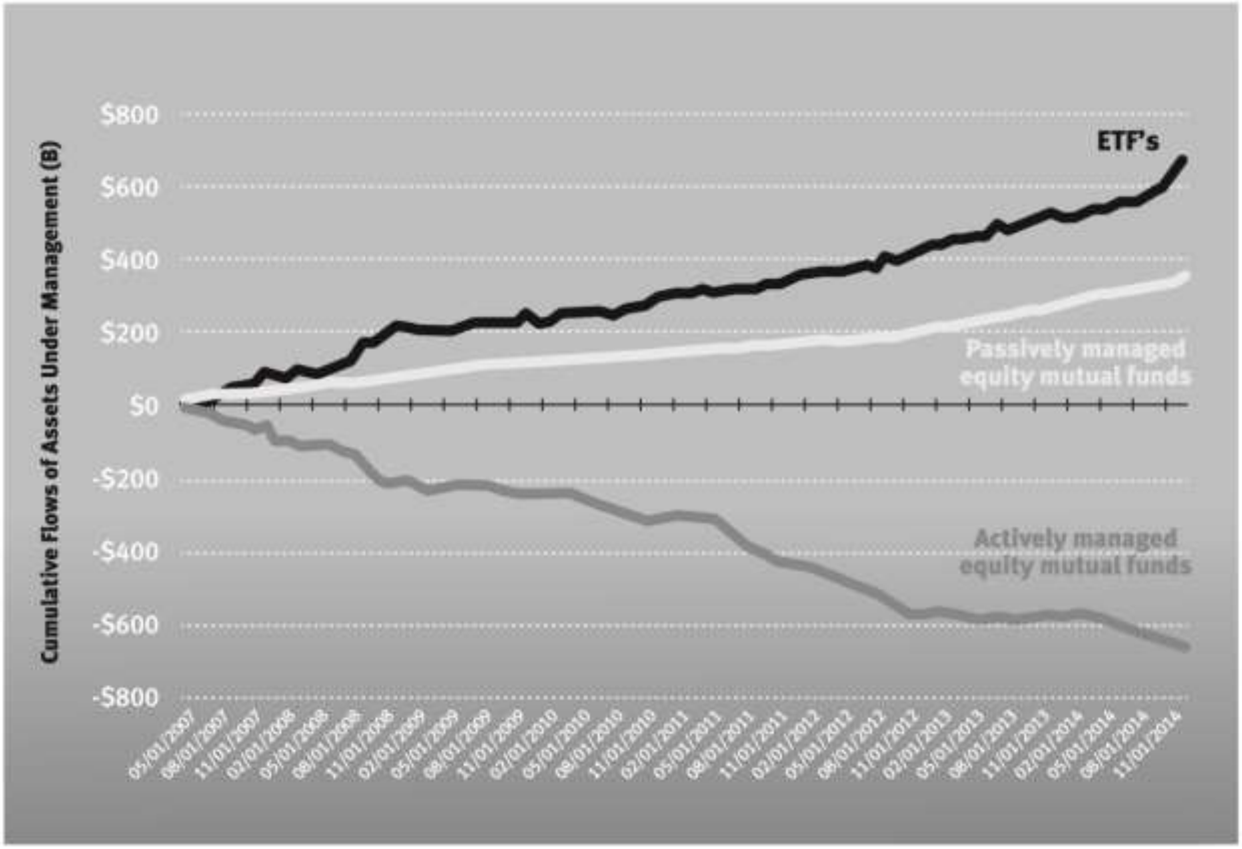
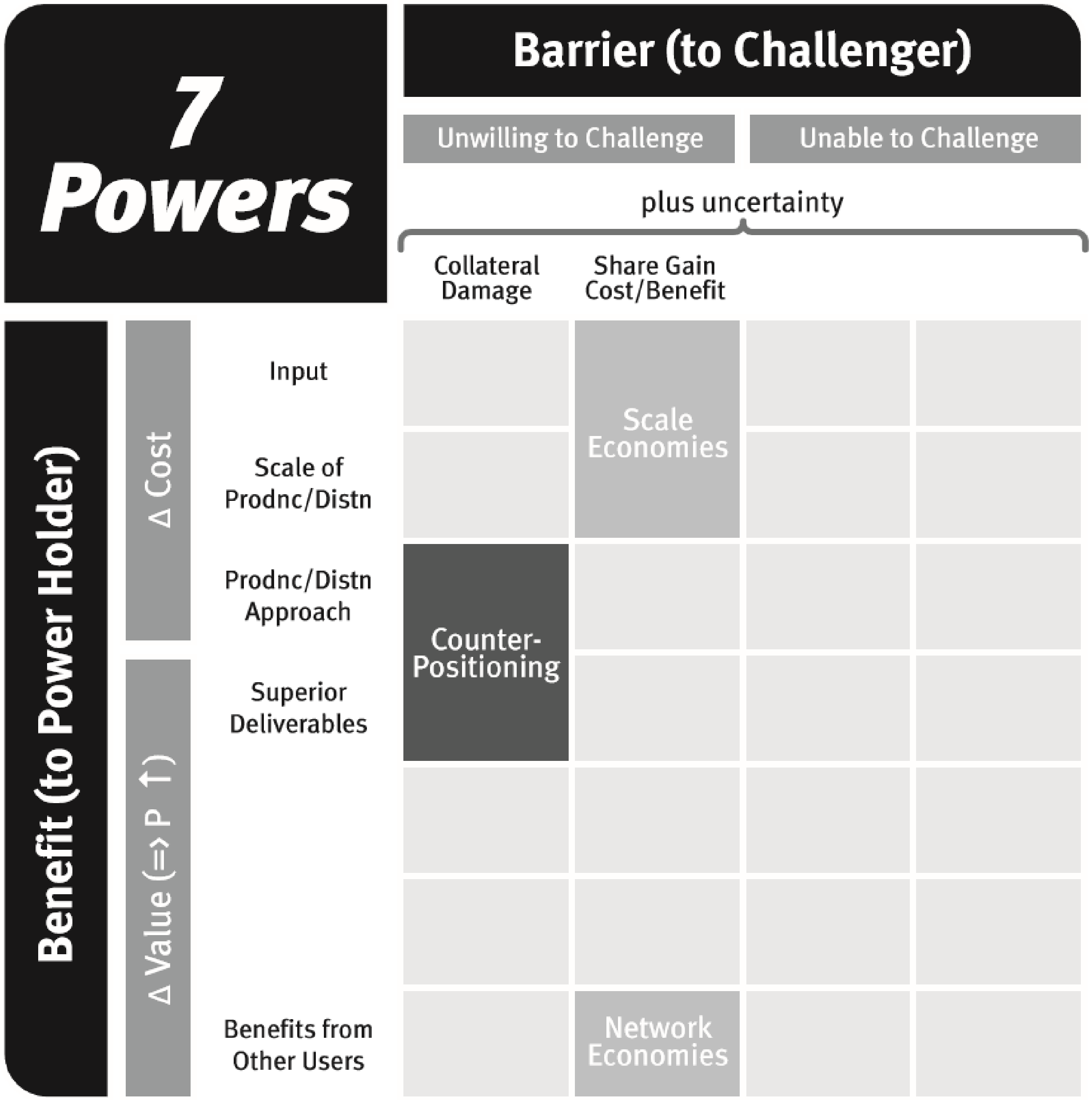


Figure 3.4: Counter-Positioning in the 7 Powers



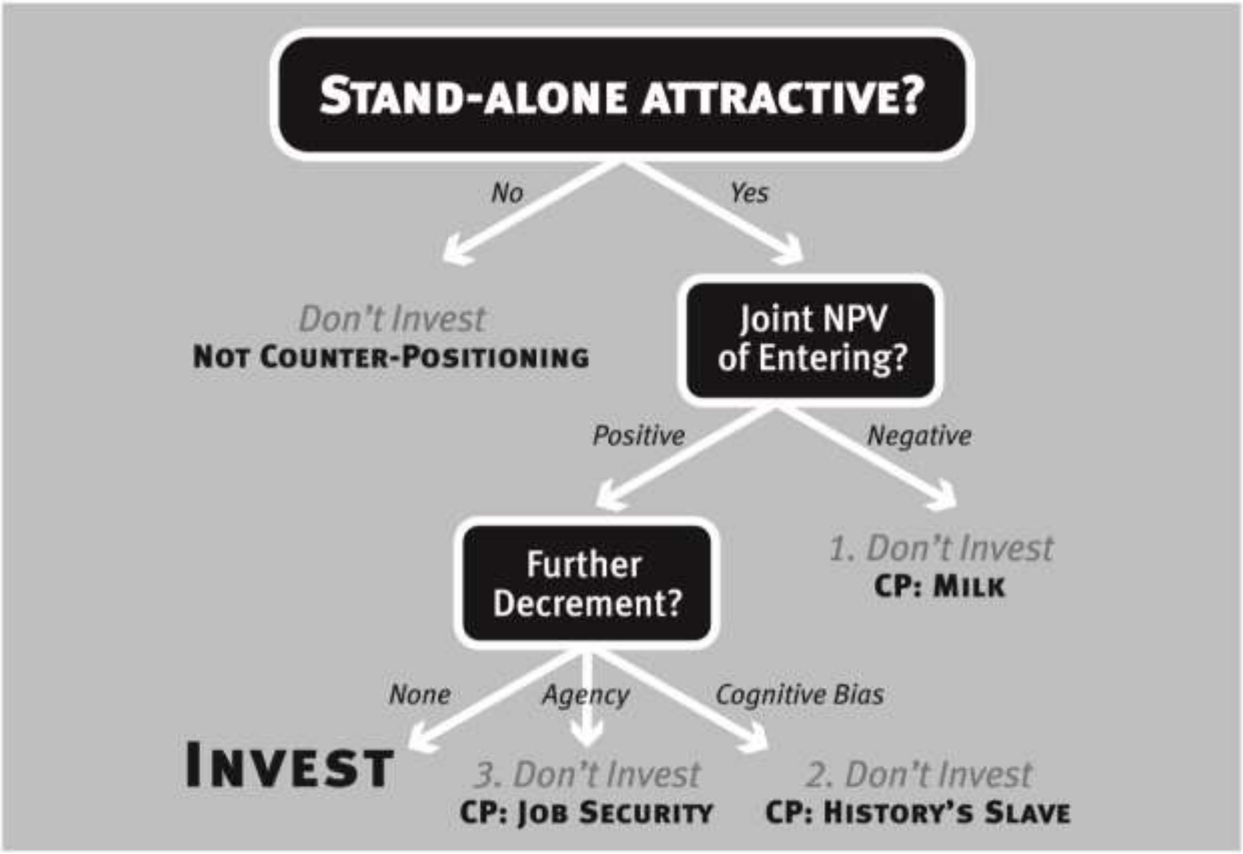

STAND-ALONE ATTRACTIVE?
No
Yes
Don't Invest
NOT COUNTER-POSITIONING
Joint NPV of Entering?
Positive
Negative
Further Decrement?
1. Don't Invest
CP: MILK
None
Agency
Cognitive Bias
INVEST
3. Don't Invest
CP: JOB SECURITY
2. Don't Invest
CP: HISTORY'S SLAVE

Figure 3.5: Power Intensity Determinants

	Industry Economics	***Competitive Position***
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Network effect intensity	Absolute difference in installed base
Counter-Positioning	New business model superiority + collateral damage to old	Binary: entrant—new model; incumbent—old model

Appendix 3.1: Derivation of Surplus Leader Margin for Counter-Positioning

To calibrate the intensity of Power, I ask the question "What governs profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?" In the case of Counter-Positioning (CP), the incumbent is W and the challenger is S.

Both business models are strictly variable cost: Profits $\equiv \pi = (P - c)\,Q$

With $P \equiv$ price per unit

$c \equiv$ variable cost per unit

$Q \equiv$ unit volume

There are two business models: Old $\equiv$ O and New $\equiv$ N

N's superior business model => ${}^{N}c < {}^{O}c$; N cannibalizes O via ${}^{N}P < {}^{O}P$.
W faces the choice of whether to enter N or not.

Surplus leader margin (SLM) is the margin the company with Power can earn while pricing is such that the margin of the weaker firm is zero. SLM is an indicator of the intensity of Power. The surplus indicated (if positive) gives S the opportunity for profits and/or Power position enforcement. In the cases of Network Economies and Scale Economies, the scale leader is S, so SLM indicates retaliatory latitude in protecting market share. In the case of CP, S is the challenger, and Power position enforcement involves diminishing the likelihood that W, the incumbent, will enter N to battle S. Such enforcement involves increasing the collateral damage.

SLM in CP will be the S's margin when the incremental profitability for W of deciding to enter N is zero.

For simplicity I will look at this as a single-period problem, although in the real world the businesses will likely be evaluating a number of periods.

So for collateral damage to just cancel out the gains to W from entering N:

Note that I will drop the W, S notation since collateral damage only refers to W's economics.

SLM => $^{N}\pi + \Delta^{O}\pi = 0$ where $\Delta^{O}\pi$ is the change in W's O business' profits *induced* by their entry into N

CP => $^{N}\text{margin} * {}^{N}\text{revenue} + {}^{O}\text{margin} * \Delta^{O}\text{revenue} = 0$

$^{N}m * [{}^{N}P * {}^{N}Q] + {}^{O}m * [{}^{O}P * \Delta^{O}Q] = 0$ where $Q \equiv$ unit volume and $m \equiv$ profit margin

$$^{N}m * [{}^{N}P * {}^{N}Q] = -\,{}^{O}m * [{}^{O}P * \Delta^{O}Q]$$

$$^{N}m = {}^{O}m * [{}^{O}P / {}^{N}P] * [-\Delta^{O}Q / {}^{N}Q]$$

Let $\delta \equiv$ W's induced cannibalization ratio of O into N: $\delta = -\Delta^{O}Q / {}^{N}Q$

So $\boxed{\mathbf{SLM = {}^{O}m * [{}^{O}P / {}^{N}P] * \delta}}$

So SLM > 0 combined with the earlier conditions of $^{N}P < {}^{O}P$ and $^{N}c < {}^{O}c$ characterize the Milk case of CP. Both a Benefit and a Barrier are evident.

Let me comment on the implications of this specification:

- If $\Delta^{O}Q = 0$. This means that W anticipates that their entry into N will cause no additional volumes losses in their base business O.
    - Then $\delta = 0$.
    - This would result in SLM = 0 so there is no CP.
    - What is going on, of course, is that there is simply no collateral damage.
    - Thus a commonly observed behavior is that Counter-Positioned incumbents will seek customer segments in which they induce no additional loss of O customers by offering N.

- For example, a *Financial Times* article of Oct. 24, 2015: *Walt Disney's most beloved characters and stories are going digital in a new streaming service that launches in the UK next month.*

  DisneyLife bundles books and music with its animated and live action films, making Disney the biggest media company yet to stream its content directly to consumers online.

  Disney will expand the service across Europe next year, with the goal of launching in France, Spain, Italy and Germany, and would add content as it becomes available, the company said.

  ...It has no plans to bring the service to the US, its biggest market, because of potential overlaps with the many agreements it has with cable and satellite companies that distribute its film and television content.

- If $\delta < 1$ (the unit gains in N are more than offset by the losses in O for W).
  - CP is unlikely. For CP, the margins would have to be attractive enough in N to offset both the lower prices in N and the volume losses.
  - Thus the incumbent would need to anticipate volume gains in N which more than offset the cannibalization of O volume induced by their entry into N.
- One of the ironies of CP is that the higher the incumbent's margins, the higher the SLM. This of course simply reflects that W has more to lose by erosion of their O business. CP therefore can present a potent challenge to an entrenched highly successful incumbent.
- The potential for cognitive bias (History's Slave) can be usefully explored by noting the elements of the SLM equation. In considering entry into N, the incumbent will often exhibit a cognitive bias that raises their expected δ, and thus increases SLM.
  - They have more certainty regarding $\Delta^{O}Q$ than ${}^{N}Q$ so they often under-

- Thus this creates a cognitive skew toward CP.

- The potential for agency effects (Job Security) can also be looked at through the lens of the SLM equation.
  - Example 1. An important decision influencer is the division head responsible for the O business.
    - The O business has been the corporation's bread and butter, so this person's voice carries a lot of weight.
    - However, the N business results are attributed to another division or group.
    - So using my example from this chapter, imagine that the active fund managers would get no credit for the assets under management in the newly formed passive funds, a quite realistic assumption.
    - This arrangement assures that ${}^{N}Q = 0$ for this individual (group), and this means $\delta = \infty$ assuring CP.
  - Example 2. At the CEO level, compensation may be set in a way that places emphasis on near-in results (this year, for example). Although I have used a single-period formulation in this appendix, the real calculation should be an NPV, and, as I have discussed, out years weigh heavily on this. The agency effects may lead to lower weights to these out years and, as I will discuss below, collateral damage tends to become less and less likely to be met as you go forward in time.

- The reader should also keep in mind that the agency and cognitive effects in CP are not mutually exclusive with the Milk case. In fact, they are frequently additive: all three effects operating at the same time.

- Dynamic effects
  - As you move forward in time, δ tends to decline, reducing Power intensity (and perhaps eliminating CP altogether).

- The reason for this is that as the aggregate cumulative cannibalization of O by N increases, ${}^{N}Q$ tends to go up because the opportunity overall for N is larger as it becomes proven and known, and $|\Delta^{O}Q|$ tends to go down as the expected loss in O business is seen to come primarily from the challenger's incursions, rather than that induced by W's entry into N.
- Also the agency and cognitive skews toward fulfilling the collateral damage condition tend to diminish over time as the uncertainty surrounding the threat of N diminishes and the agents aligned with O in W tend to lose credibility and influence.
- Since δ tends to decline, SLM declines and the collateral damage may be insufficient to deter W's entry into N. This is the capitulation point mentioned in the chapter.

- I understand that this specification is highly stylized. Even so, the future profit calculations in corporations tend not to be theoretically complex, so even this stylized representation may capture much of what is going on.
- Tactically, it is probably a good idea for S to set prices at first such that ${}^{N}m$ is very low—much lower than ${}^{O}m$.
  - ${}^{N}m$ is usually observable by W as is ${}^{N}P$, whereas δ is not.
  - So if $[{}^{N}P/{}^{O}P]$ is quite small, then W has to be quite optimistic about a low cannibalization rate (δ) to lead to an SLM > 0 thus creating CP.
  - A special case is one in which S offers N at first for ${}^{N}P$ such that ${}^{N}m$ < 0. This makes $[{}^{N}m/{}^{O}m] < 0$ and assures the collateral damage condition is met for the periods of this pricing. Since ${}^{N}P$ is observed but S's motivation is not, W may well discount the possibility that S will eventually raise prices such that ${}^{N}m > 0$ whereas S can know this, assuming they are a price leader.

Figure 4.1: SAP Stock Price[38]

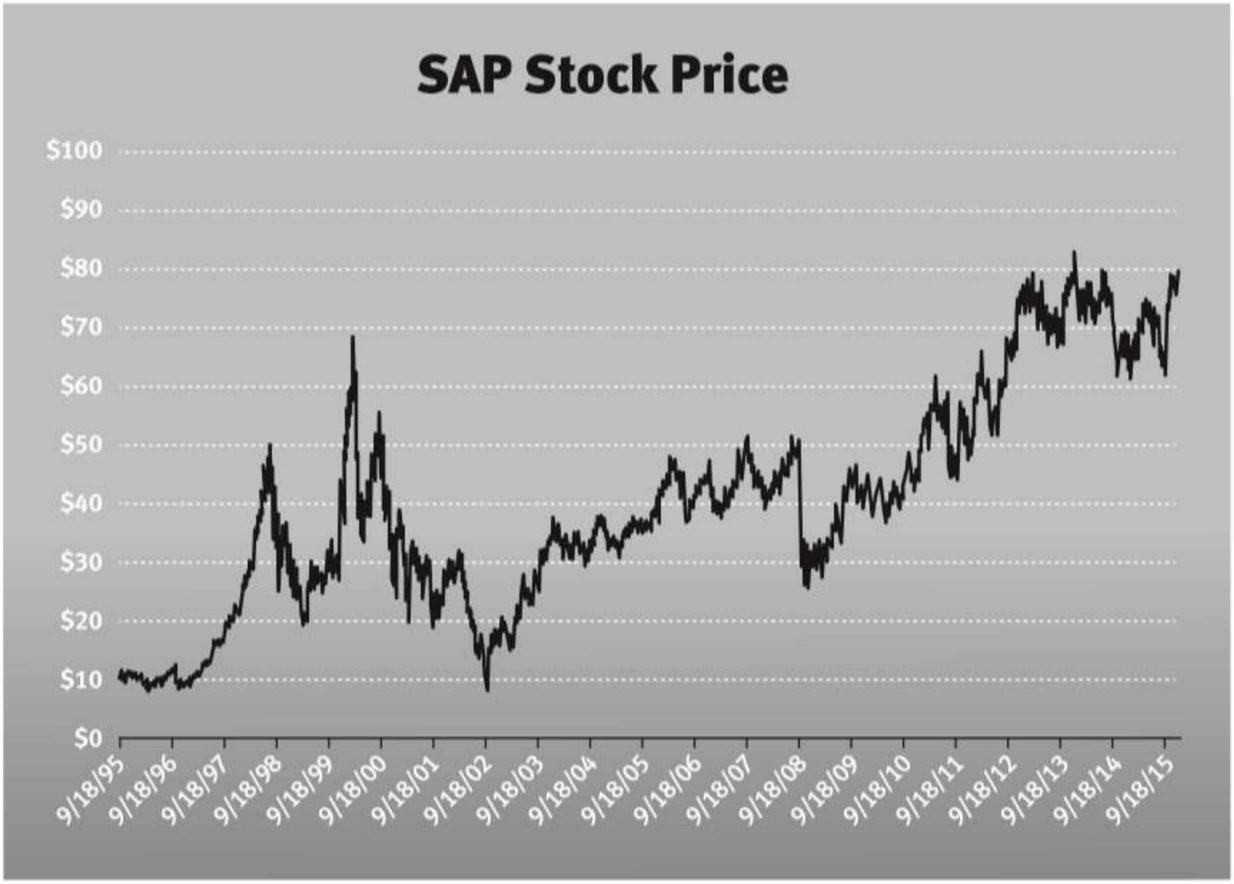


Figure 4.2: Switching Costs in the 7 Powers

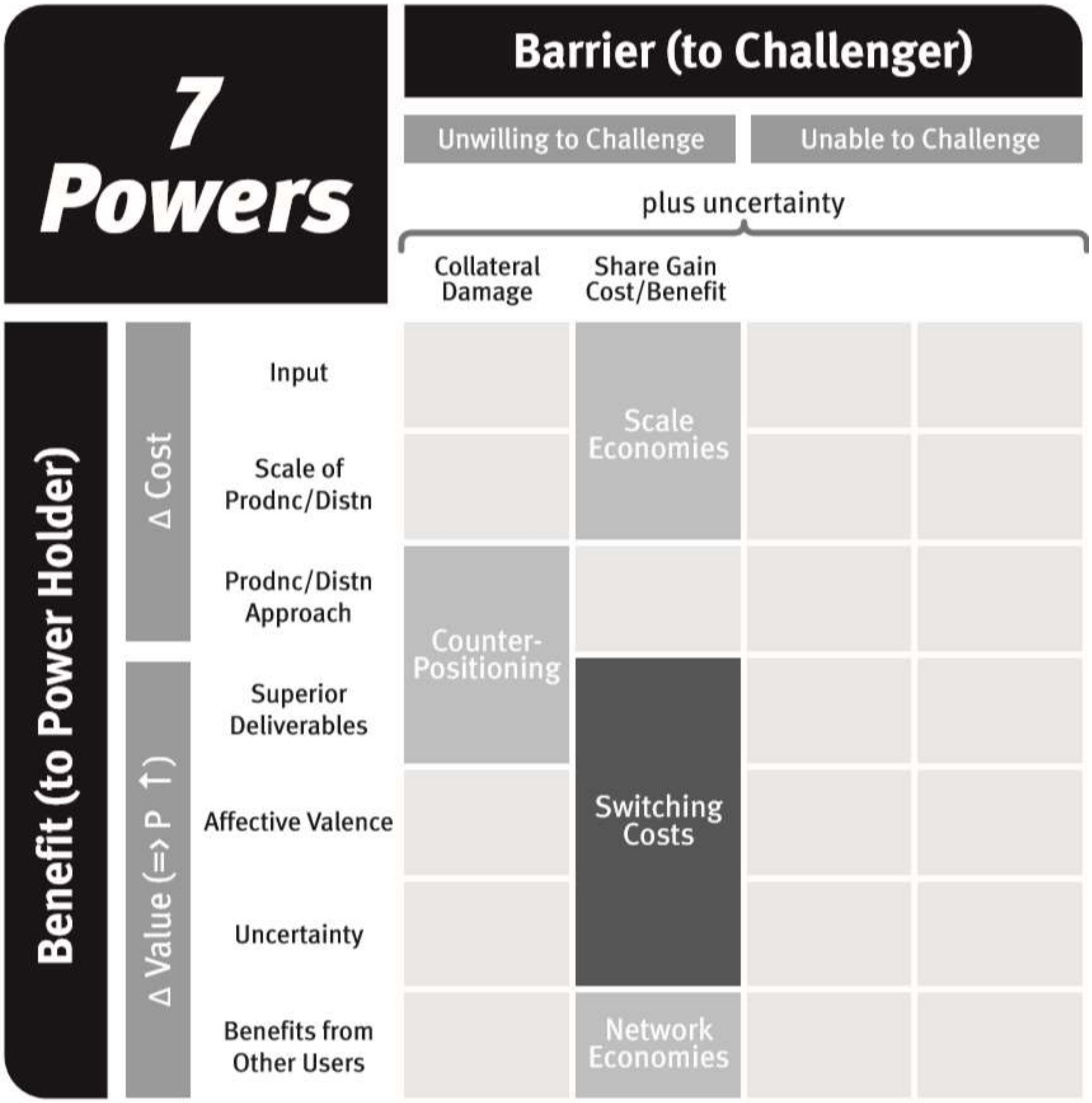


Figure 4.3: SAP Product Offerings[46]

SAP Advanced Planner and Optimizer (APO)

SAP Analytics

SAP Advanced Business Application Programming (ABAP)

SAP Apparel and Footwear Solution (AFS)

SAP Business Information Warehouse (BW)

SAP Business Intelligence (BI)

SAP Catalog Content Management ()

SAP Convergent Charging (CC)

SAP Enterprise Buyer Professional (EBP)

SAP Enterprise Learning

SAP Portal (EP)

SAP Exchange Infrastructure (XI) (From release 7.0 onwards, SAP XI has been renamed as SAP Process Integration [SAP PI])

SAP Extended Warehouse Management (EWM)

SAP GRC (Governance, Risk and Compliance)

SAP EHSM (Environment Health Safety Management)

Enterprise Central Component (ECC)

SAP HANA (formerly known as High-performance Analytics Appliance)

SAP Human Resource Management Systems (HRMS)

SAP Success Factors

SAP Internet Transaction Server (ITS)

SAP Incentive and Commission Management (ICM)

SAP Knowledge Warehouse (KW)

SAP Manufacturing

SAP Master Data Management MDM)

SAP Rapid Deployment Solutions (RDS)

SAP Service and Asset Management

SAP Solutions for mobile businesses

SAP Solution Composer

SAP Strategic Enterprise Management (SEM)

SAP Test Data Migration Server (TDMS)

SAP Training and Event Management (TEM)

SAP NetWeaver Application Server (Web AS)

SAP xApps

SAP Supply Chain Performance Management (SCPM)

SAP Sustainability Performance Management (SUPM)

Figure 4.4: Number of SAP Acquisitions by Year[48]

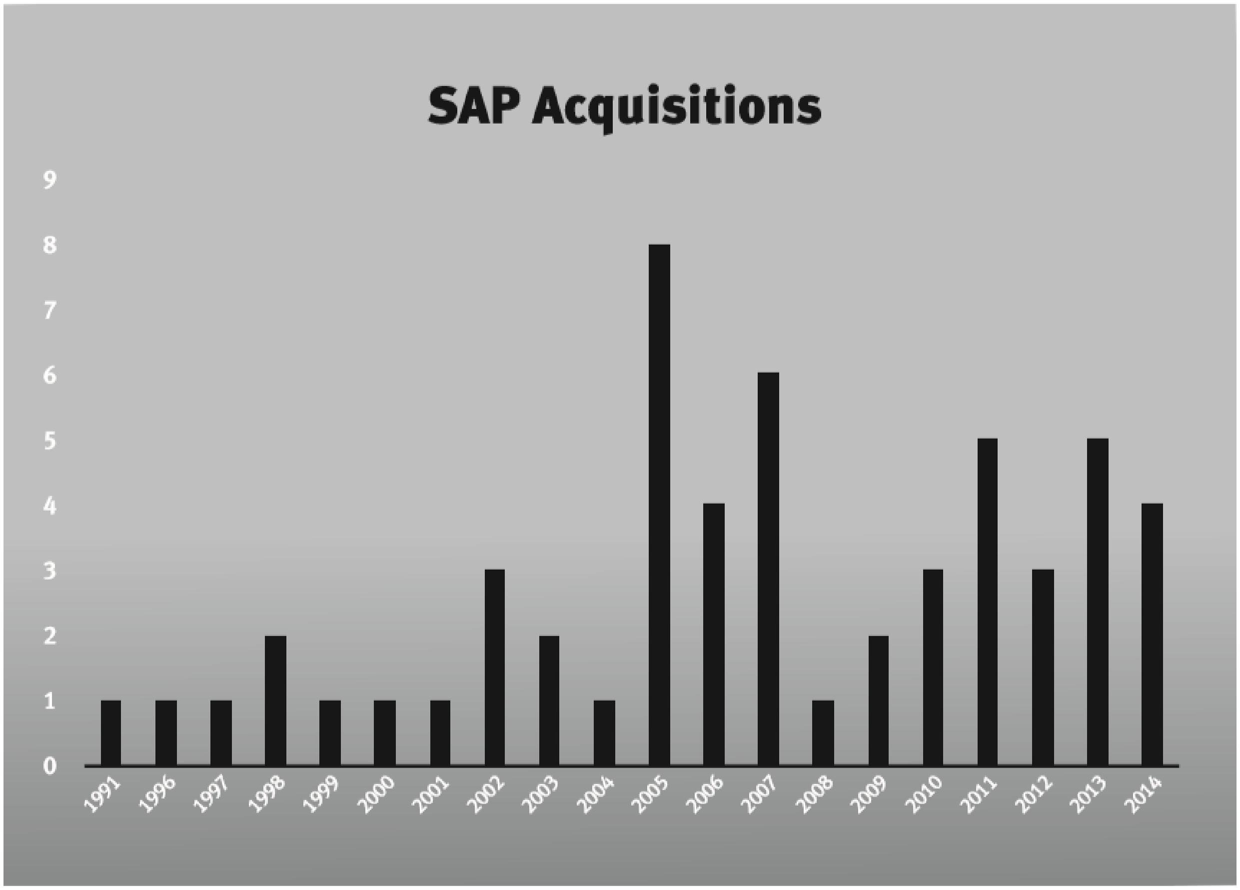


Figure 4.5: Power Intensity Determinants

	Industry Economics	Competitive Position
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Intensity of network effect	Absolute difference in installed base
Counter-Positioning	New business model superiority + collateral damage to old	Binary: entrant—new model; incumbent—old model
Switching Costs	Magnitude (intensity) of switching costs	Number of current customers

Appendix 4.1: Surplus Leader Margin for Switching Costs

S is the strong company and W the weak company. In this case the "weak company" is the company not having the customer.

${}_{S}Q$ consumers have already adopted S's product. I will now examine the benefit that accrues to S due to sales of subsequent products to ${}_{S}Q$.

Suppose for simplicity that the utility of the subsequent product is the same for both firms. S is able to charge a price premium due to switching costs Δ:

$${}_{S}P = \Delta + {}_{W}P$$

Also for simplicity, assume that there are no fixed costs to production.

$$\text{Profit} \equiv \pi = [P - c]\ Q$$

with $P \equiv$ price

$c \equiv$ variable cost per unit

$Q \equiv$ units produced per time period

As an indication of leverage, assess:

What governs S's margins if P is set $\ni {}_{W}\pi = 0$?

$${}_{W}\pi = 0 \Rightarrow \quad 0 = ({}_{W}P - c)\ {}_{S}Q \quad \Rightarrow {}_{W}P = c$$

Δ is the Switching Cost per unit

S can charge a premium, so ${}_{S}P = \Delta + c$

$$\therefore \quad {}_{S}\pi = [(\Delta + c) - c]\ {}_{S}Q$$

$${}_{S}\pi = \Delta\ {}_{S}Q$$

$$\boxed{\textbf{SLM} = \boldsymbol{\Delta}}$$

Industry Economics: Δ

Competitive Position: ${}_{S}Q$

Figure 5.1: Price Comparisons for Engagement Rings

Seller and Product, 1 carat	Starting Price in US Dollars
Tiffany: The Tiffany Setting (I VS2)	$12,000
Cartier: Solitaire 1895 (H VS2)	$14,800
DeBeers: Signature	$12,200
Blue Nile: Classic 6-prong (I VS2)	$6,697

#54
12-03-2009, 8:43PM

User X
Location: Encinitas, CA

I bought Tiffany and knew I was getting soaked. Didn't matter — happy then and would do it again (in fact, I upgraded her wedding band to matching channel diamonds years later.)

My priority was to buy the best from the best without any doubt of quality/certification/etc. Size was not important. I wanted indisputable perfect to match her. We're not showy people and have never played up the fact the rings are from Tiffany's. It was more appropriate to me to buy a reasonalble size stone and know with quiet confidence that the ring is timeless — not a cheap knock-off or guady bauble.

The other thing that I was mindful of was that someday (like when I'm dead) one of my grandchildren will inherit it. Part of the rationale for me was "instant heirloom." One of these future grandkids is going to think "Damn, grandpa was ****ing cool!"

Last edited by User X; 12-03-2009 at 8:46 PM.

Posted 11 January 2007 — 8:38 PM

User X

In trying to save some money and get more bang for my buck, I chose to purchase an engagement ring from a so-called reputable jeweler years ago and got a lemon (the jeweler is no longer in business). I now find myself spending 10 times more for a replacement ring from Tiffany's. To me, the expense is well worth the peace of mind and the expression on my wife's face was priceless. A diamond from John Doe Jewelers or Costco does not have the same effect as one from Tiffany's. The experience alone makes it worth the expense.

Figure 5.1: Completed eBay Auction for Tiffany Box

Genuine Tiffany & Co. Empty Engagement Ring Box, Bag, Tiffany & Co

Item condition: **Pre-owned**

Ended: May 27, 2015, 12:22PM

Winning bid: **US $122.50** [33 bids]

Located in United States

Seller information

100% Positive feedback

Follow this seller

See other items

Figure 5.2: Annual Profit Margins for Blue Nile and Tiffany[51]

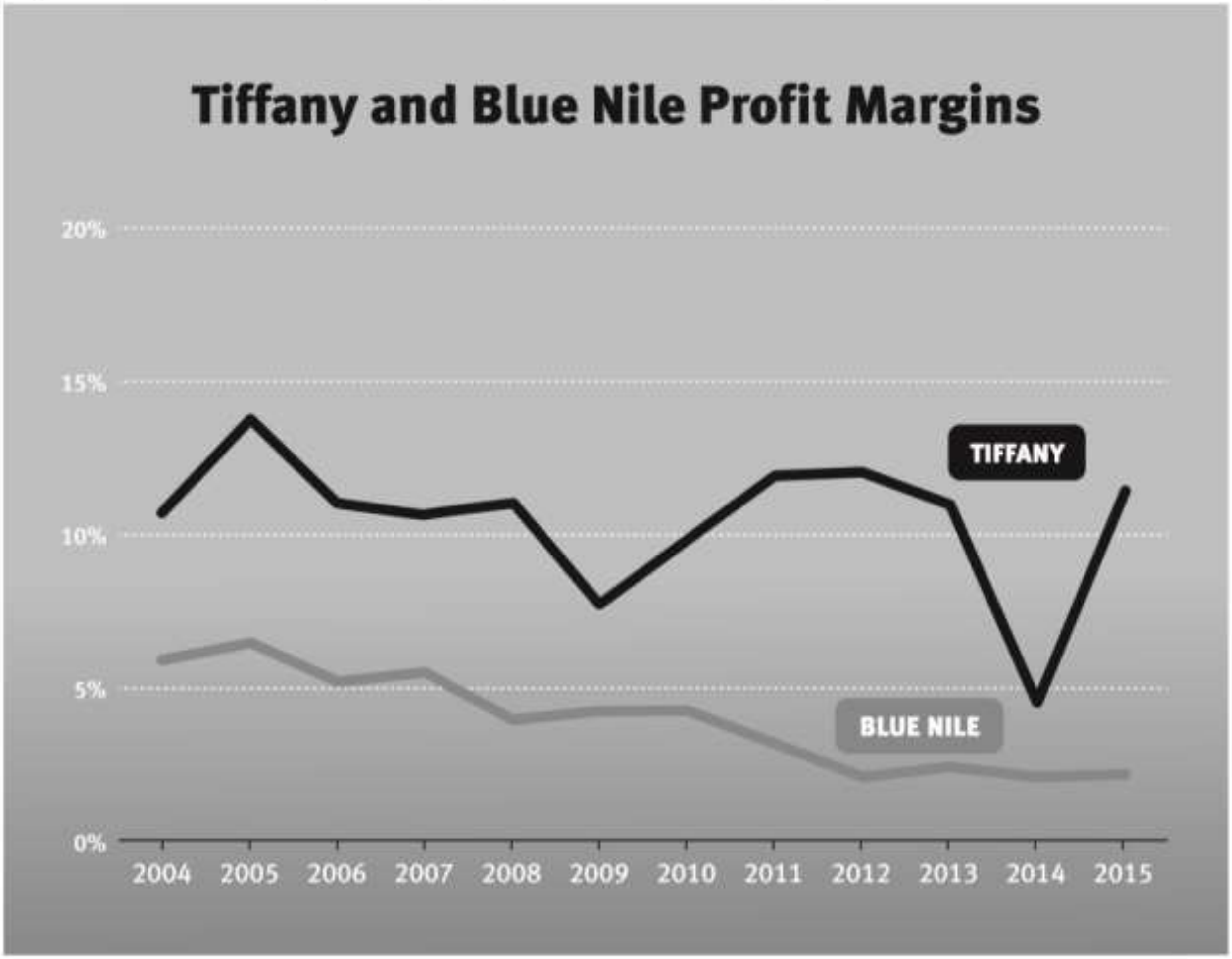


Figure 5.3: Tiffany Stock Price[52]

Figure 5.4: Branding in the 7 Powers

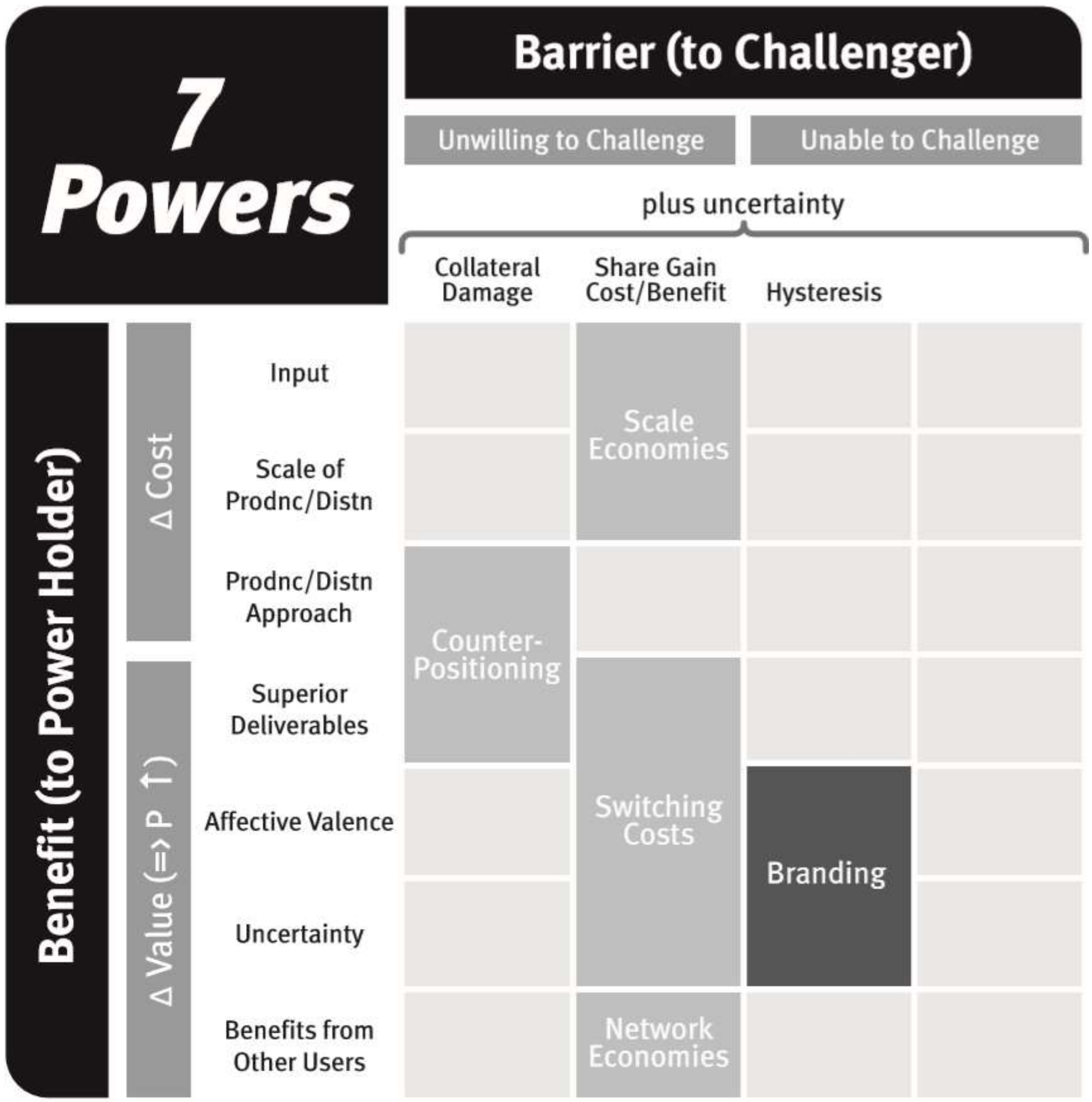


Figure 5.5: Power Intensity Determinants

	Industry Economics	***Competitive Position***
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Intensity of network effect	Absolute difference in installed base
Counter-Positioning	New business model superiority + collateral damage to old	Binary: entrant—new model; incumbent—old model
Switching Costs	Magnitude (intensity) of switching costs	Number of current customers
Branding	Time constant and potential magnitude of Branding effect	Duration of brand investing

Appendix 5.1: Surplus Leader Margin for Branding

To calibrate the intensity of Power, I ask the question "What governs profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?"

S is the strong company (the one with Branding Power) and W the weak company.

To derive a formula for SLM for Branding, I need to specify what determines the upper envelope of the price premium enjoyed by the strong firm (S). B(t) is that specification.

$$B(t) = Z/(1 + (z\text{-}1)e^{-Ft}) * D_t * U_t$$

- $B(t)$ = branding price multiple at time t
- Z = maximum potential branding multiple for this good type, $Z > 2$
- F = brand cycle time compression factor, $F > 0$
- D_t = brand dilution at time t, $0 \leq D \leq 1$
- U_t = brand underinvestment at time t $0 \leq U \leq 1$

B(t) is an increasing function of t, reflecting the reality that Branding requires action over time to be increased. The logistic function was chosen to reflect the reinforcing aspect of Branding investment, while allowing diminishing marginal returns over time. The particular form specified above for B(t) ensures that $B(t)=1$ at $t=0$ by adjusting the location parameter as a function of F and Z. When F is larger, the logistic curve steepens and the brand cycle time is shorter. When F is smaller, the logistic curve grows shallower and the brand cycle time is longer. As seen in Figure 5.6, $D = 1$ if there is no brand dilution, and $U = 1$ if there is no brand underinvestment; otherwise, D and U reduce the brand multiple in a given period by some fraction.

Time determines competitive position, because it determines the ability of a competitor starting at time $t = 0$ to catch up to the strong firm at time $t = t$. Z, and F (i.e., B()) determines industry position. As seen in Figure 5.6, the weak competitor falls further and further behind as the length of the life cycle grows;

so too then does it become harder and harder for that weak competitor to catch up to the strong firm. The sustainability of the brand depends on the shape of the B(t) function relative to B(0) for all t in the case of the weakest competitor with no Branding, and an alternative B'(t) function (i.e. an alternative F') and alternative t for another competitor.

For simplicity, assume that there are no fixed costs to production.

$$\text{Profit} \equiv \pi = [P - c]\, Q$$

with $P \equiv$ price

$c \equiv$ marginal cost per unit

$Q \equiv$ units produced per time period

As an indication of leverage, assess:

What governs S's margins if P is set $\ni {}_W\pi = 0$?

$${}_W\pi = 0 \Rightarrow \qquad 0 = ({}_WP - c)\, {}_SQ \Rightarrow {}_WP = c$$

S can charge a premium as a multiple of ${}_WP$, so ${}_SP = B(t)^*c$

$$\therefore \qquad {}_S\pi = [B(t)^*c - c]\, {}_SQ = [(B(t)\text{-}1)^*c]\, {}_SQ$$

$${}_S\pi = (B(t)\text{-}1)c\, {}_SQ$$

$${}_S\text{Margin} = (B(t)\text{-}1)c\, {}_SQ/(B(t)c\, {}_SQ) = 1 - 1/B(t)$$

$$\text{So} \quad \boxed{\mathbf{SLM = 1 - 1/B(t)}}$$

where the function B() represents the industry economics defining the magnitude and sustainability of leverage through Z and F, respectively, and t represents the competitive position that S has relative to W in developing Branding Power.

Figure 5.6: Branding as a Function of Time, with Different Compression Factors

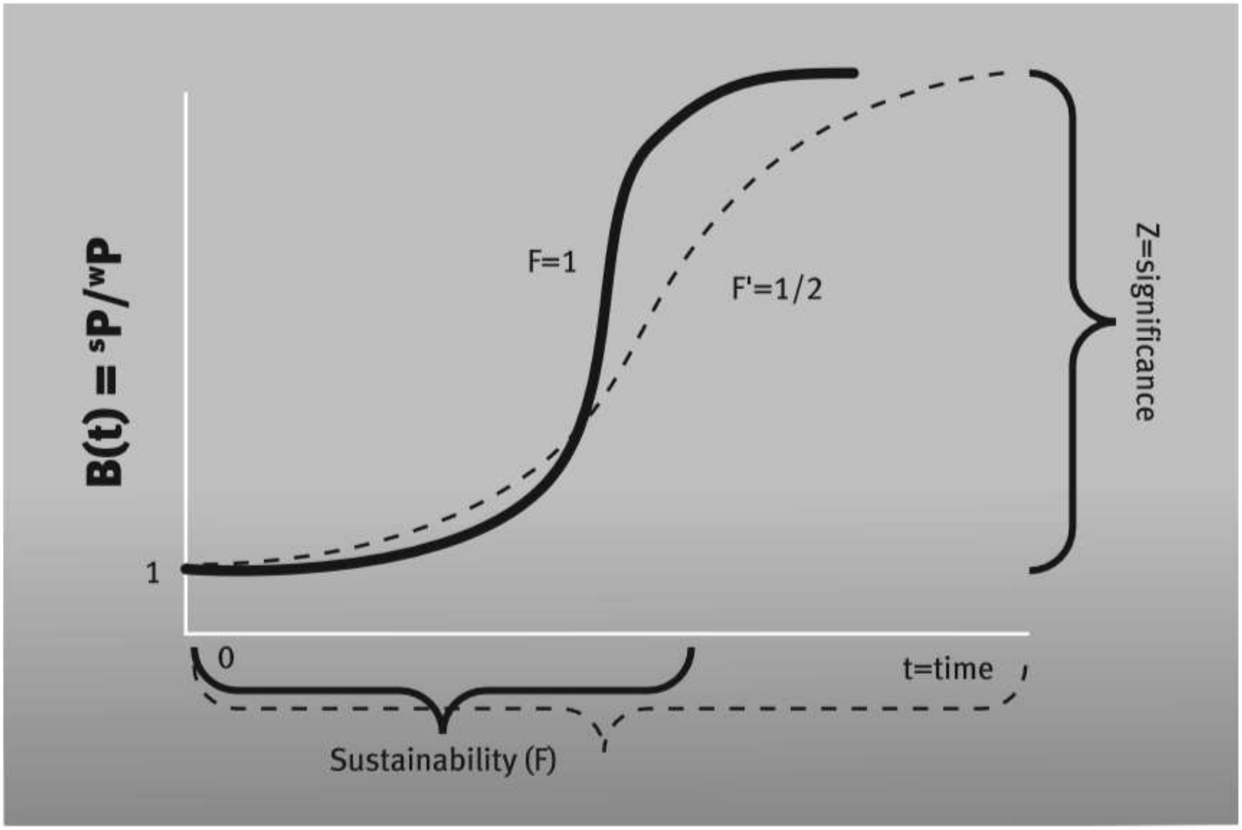
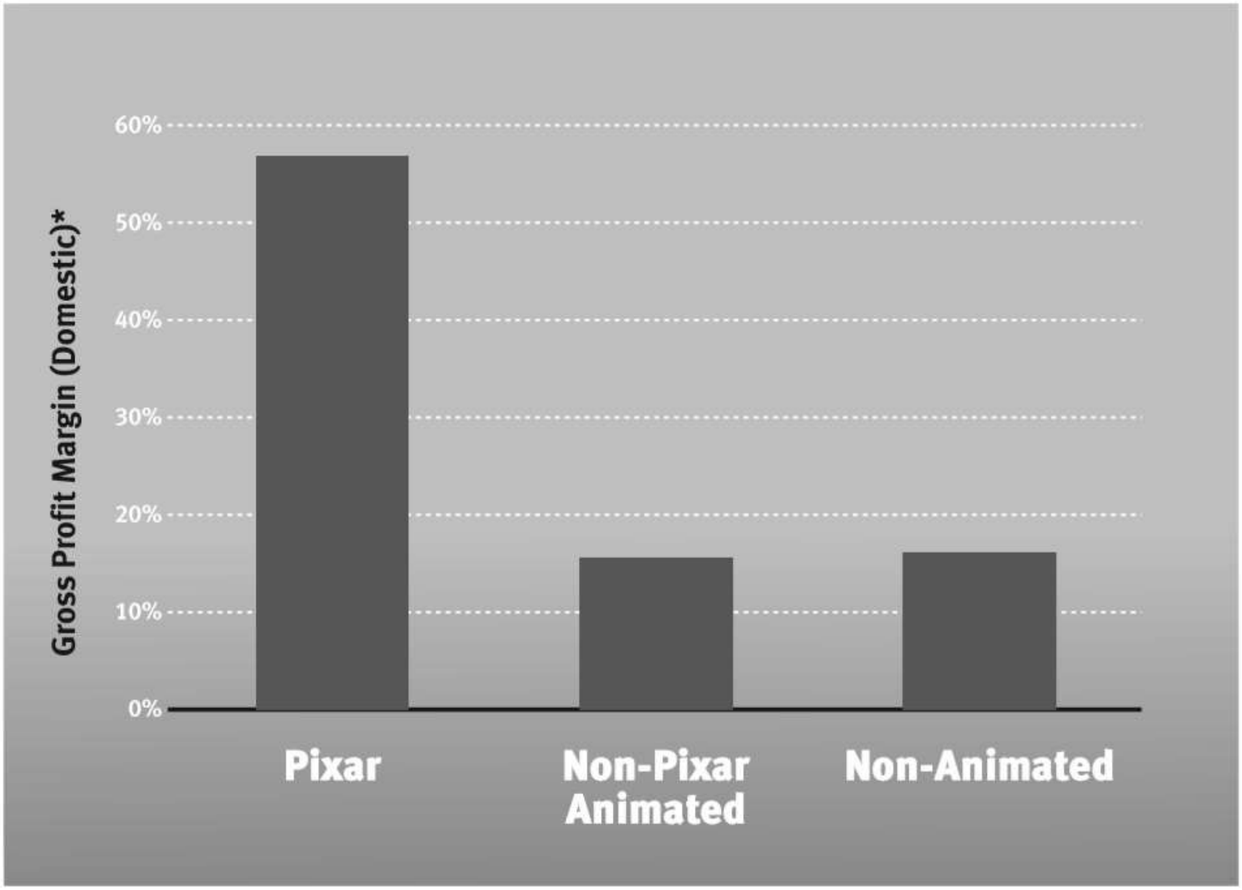


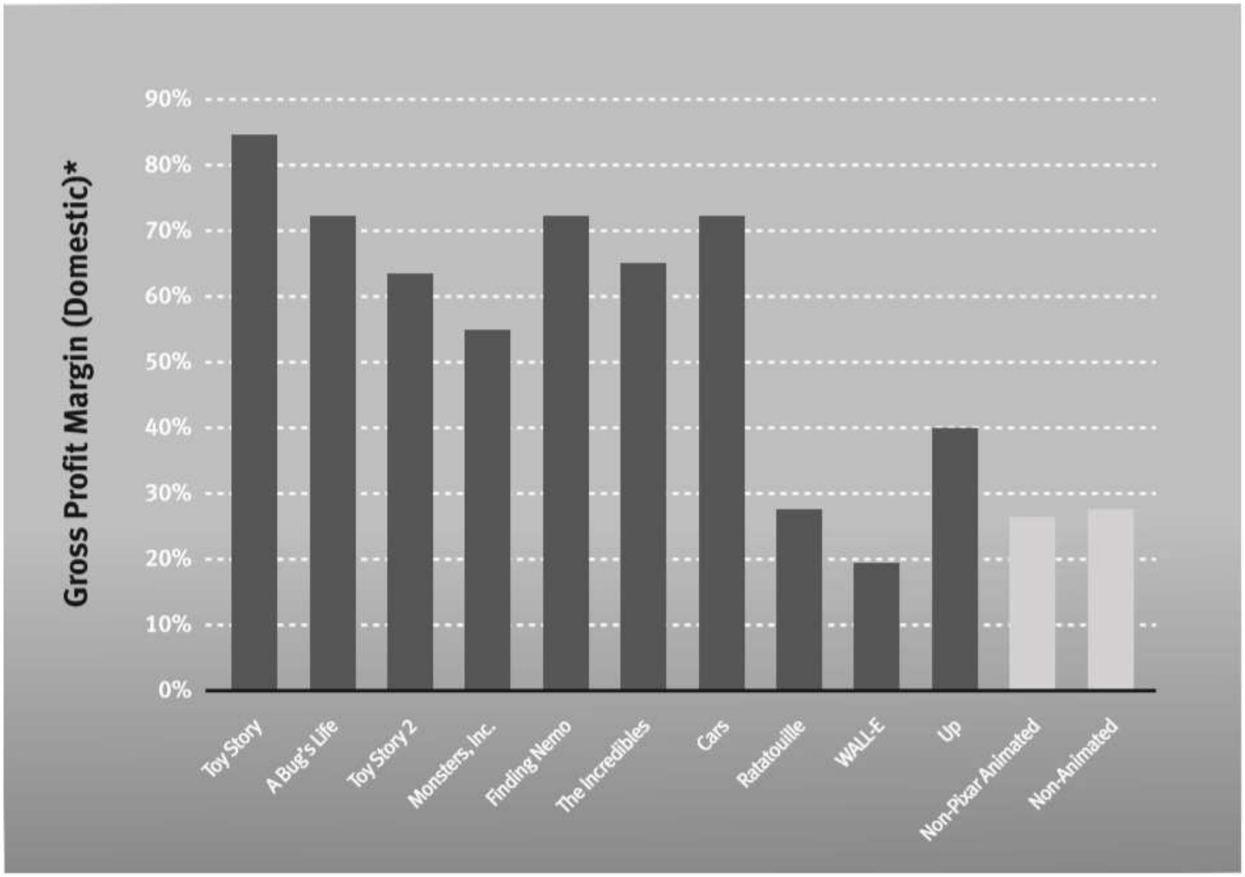
Figure 6.1: The First Ten Pixar Films

Figure 6.2: US Gross Profit Margin of Theatrical Release Movies (1980–2008)[56]



*(US Box Office – Production Costs)/(Production Costs)

Figure 6.3: Gross Profit Margin of Pixar Films vs. Industry Averages (1980–2008)



*(US Box Office – Production Costs)/(Production Costs)

Figure 6.4: Cornered Resource in the 7 Powers

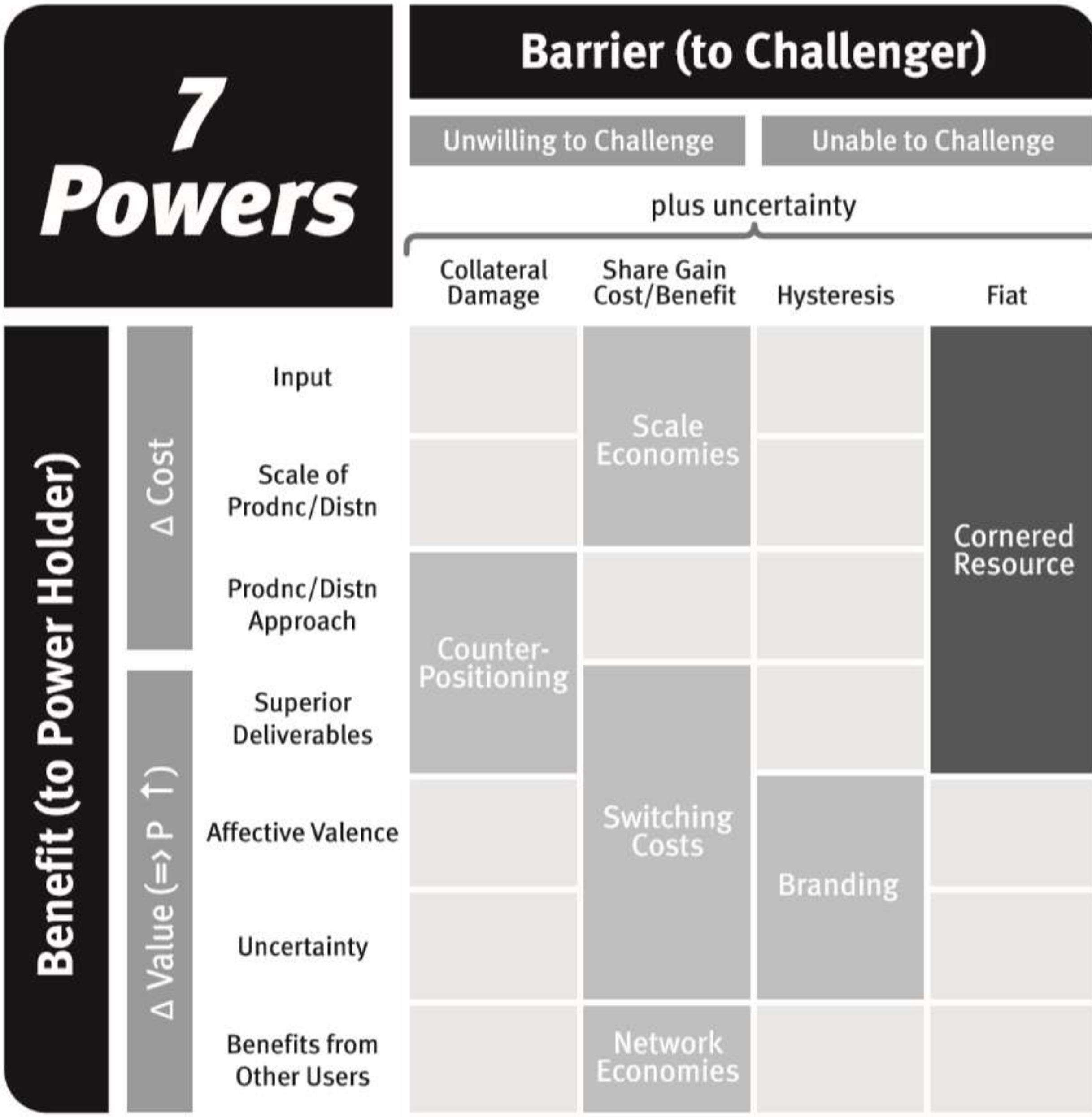


Figure 6.5: Early Pixar Film Directors

Movie	***Year***	***Director***	***Original Toy Story Team?***
Toy Story	1995	John Lasseter	
A Bug's Life	1998	John Lasseter	
Toy Story 2	1999	John Lasseter	
Monsters, Inc.	2001	Pete Docter	
Finding Nemo	2003	Andrew Stanton	
The Incredibles	2004	Brad Bird	
Cars	2006	John Lasseter	
Ratatouille	2007	Brad Bird	
WALL-E	2008	Andrew Stanton	
Up	2009	Pete Docter	
Toy Story 3	2010	Lee Unkrich	

Figure 6.6: Power Intensity Determinants

	Industry Economics	***Competitive Position***
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Network effect intensity	Absolute difference in installed base
Counter-Positioning	New business model superiority + collateral damage to old	Binary: entrant—new model; incumbent—old model
Switching Costs	Magnitude (intensity) of switching costs	Number of current customers
Branding	Time constant and potential magnitude of Branding effect	Duration of brand investing
Cornered Resource	Price &/or cost increment due to CR	Preferred access at a non-arbitraging price

Appendix 6.1: the Resource Based View

The notion of resource is a broad and inclusive one that goes far beyond the considerations of this chapter. There is a major school of thought in Strategy, the Resource Based View ("RBV"), which focuses on resources. I have benefited from the fine scholarship of the RBV and even had the privilege of studying under the brilliant Professor Richard R. Nelson, one of the RBV pioneers.

Businesses encompass not just products and services, but the abilities that enable their efficient production. There are immediate abilities, specific to current output, and higher-level abilities, which circumscribe the company's domains of competitiveness. There are even further stages beyond these which shape the ways in which these higher-level abilities may transform over time. Core competencies, distinctive competencies, routines, capabilities and dynamic capabilities all figure into this conversation.

I have intentionally restricted this chapter's treatment of resources. First, my treatment narrows the topic by looking only at resources which qualify as Power. The tests above are meant to help the practitioner eliminate resources that may well be notable but are not strategic in the sense that I use the term.

Secondly, though, a more profound narrowing: this part of my book is restricted to statics of Power. The RBV figures more heavily into dynamics. When I move on to that topic in Part II, we will observe that invention is the first cause of Power. There, in exploring the endogenous determinants of invention, the concept of resources, in the broader sense used by the RBV, will feature more prominently.

Appendix 6.2: Surplus Leader Margin for Cornered Resource

To calibrate the intensity of Power, I ask the question "What governs profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?"

Suppose the Cornered Resource (CR) possessed by S results in superior deliverables. This, for example, fits the Pixar case of consistently compelling movies.

We further suppose that S realizes a per unit increase in profits from this resource of Δ. This could come from a price increase due to superior deliverables or a cost decrease. (In the Pixar case, the superior deliverables are more focused on getting higher volumes, theater attendance. One way to think about this: the Δ that would result if pricing were such that only average volumes resulted).

Also for simplicity, assume that there are no fixed costs to production.

Profit $\equiv \pi = [P - c + \Delta]\,Q$ with $P \equiv$ price

$c \equiv$ variable cost per unit

$Q \equiv$ units produced per time period

Also suppose that the incremental cost of the CR is the fixed amount per year k. In the Pixar case this would be the extra annual compensation that would be paid to the discussed core group *above* what would be paid, if individuals with similar training were hired to replace them.

So S' profit $\equiv {}_S\pi = [{}_WP + \Delta - c]\,{}_SQ - k$

For the Surplus Leader Margin (SLM) we assess:

What governs S's margins if P is set $\ni {}_W\pi = 0$?

${}_W\pi = 0 \Rightarrow \quad 0 = ({}_WP - c)\,{}_SQ \Rightarrow {}_WP = c$

S has an additional profit from the CR: Δ

S also incurs an additional fixed cost of k (k need not be positive)

So ${}_S\pi = [(\Delta + {}_WP) - c]\,{}_SQ - k$ substituting ${}_WP = c$

${}_S\pi = \Delta\,{}_SQ - k$ dividing by revenues $[(\Delta + {}_WP)*{}_SQ]$ results in margin

So $\quad {}_{S}\text{margin} = \Delta\ {}_{S}Q/(\Delta + {}_{W}P)^{*}{}_{S}Q - k/(\Delta + {}_{W}P\ )^{*}{}_{S}Q$

$$\boxed{\mathbf{SLM = \Delta/(\Delta + {}_{W}P) - k/(\Delta + {}_{W}P)^{*}{}_{S}Q}}$$

= [Margin increase due to CR] – [CR cost per dollar sales]
Industry economics: Δ, k
Competitive position: control of CR by fiat or not

Figure 7.1: Shares in US Automobile Market[65]

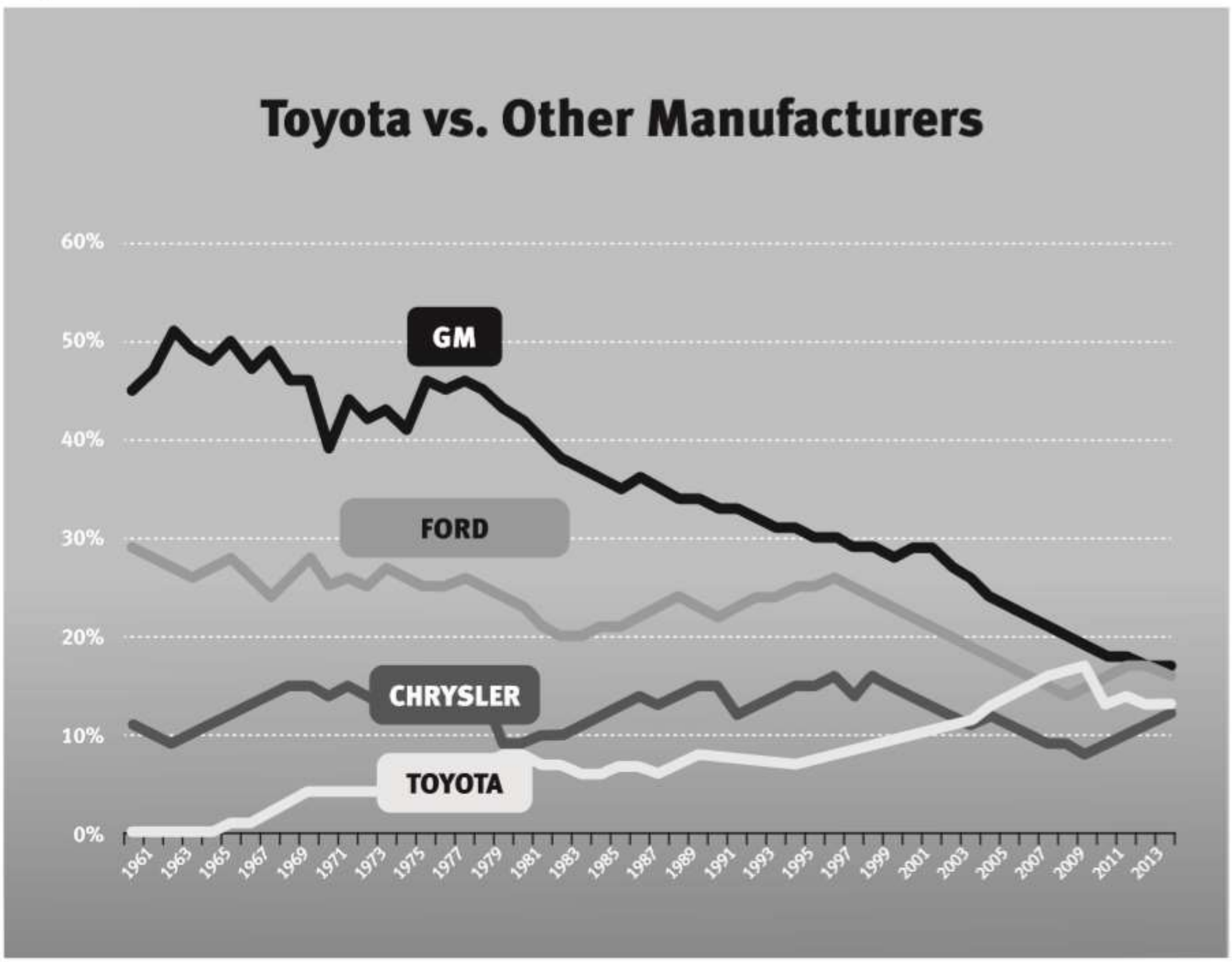


Figure 7.2: Toyota Stock Price in US Dollars[68]

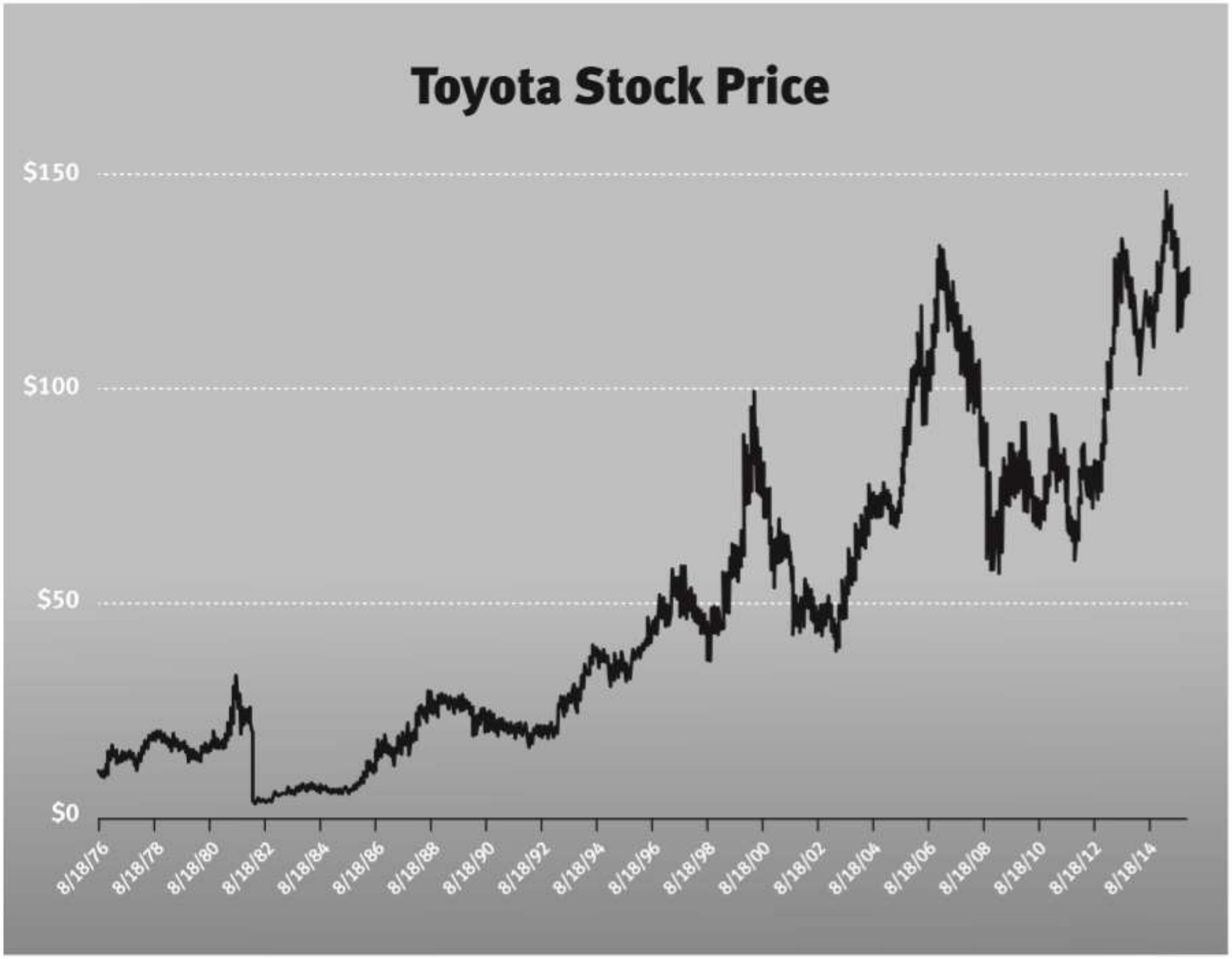


Figure 7.3: Process Power in the 7 Powers

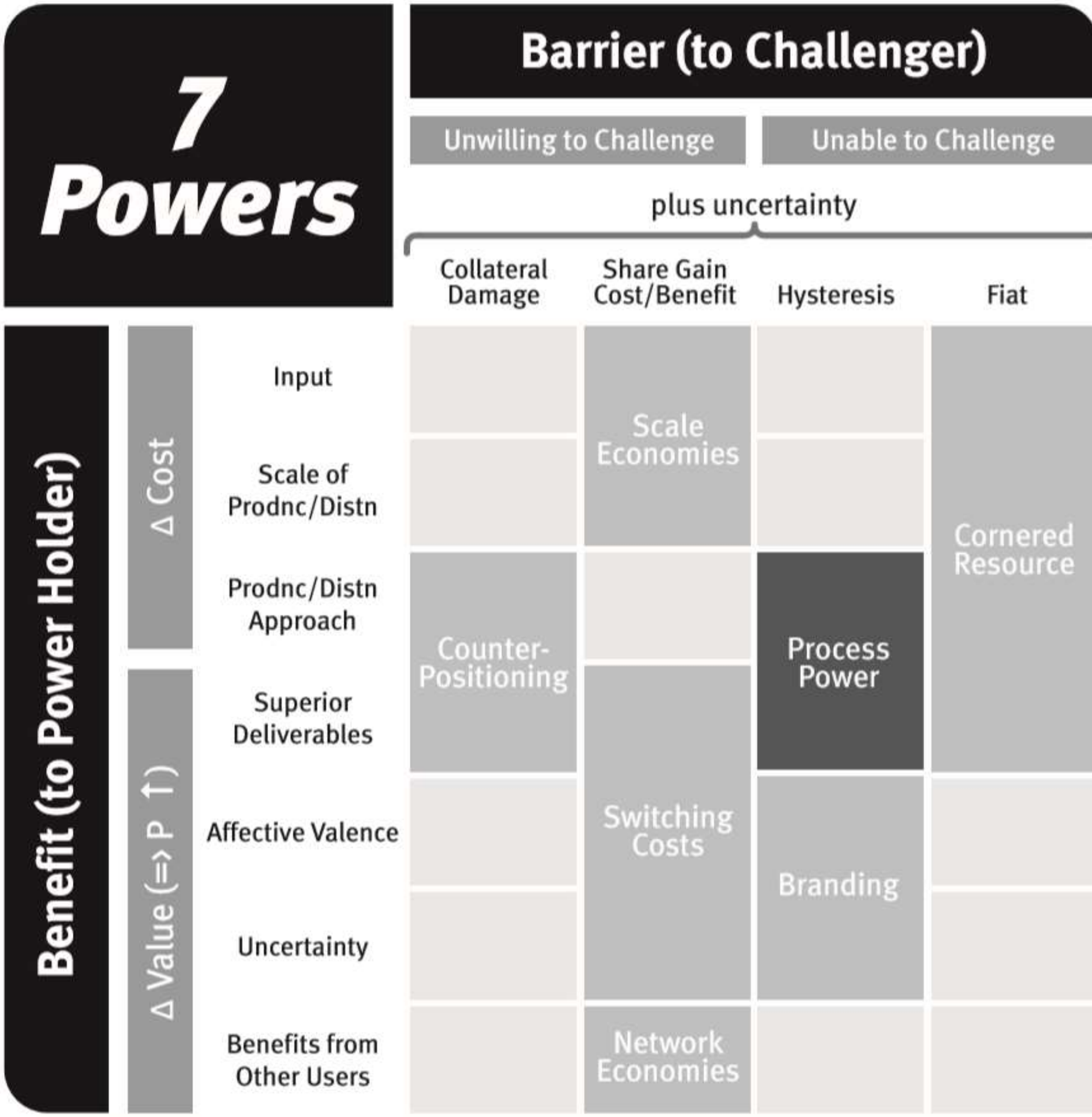


Figure 7.4: Experience Curve Sample

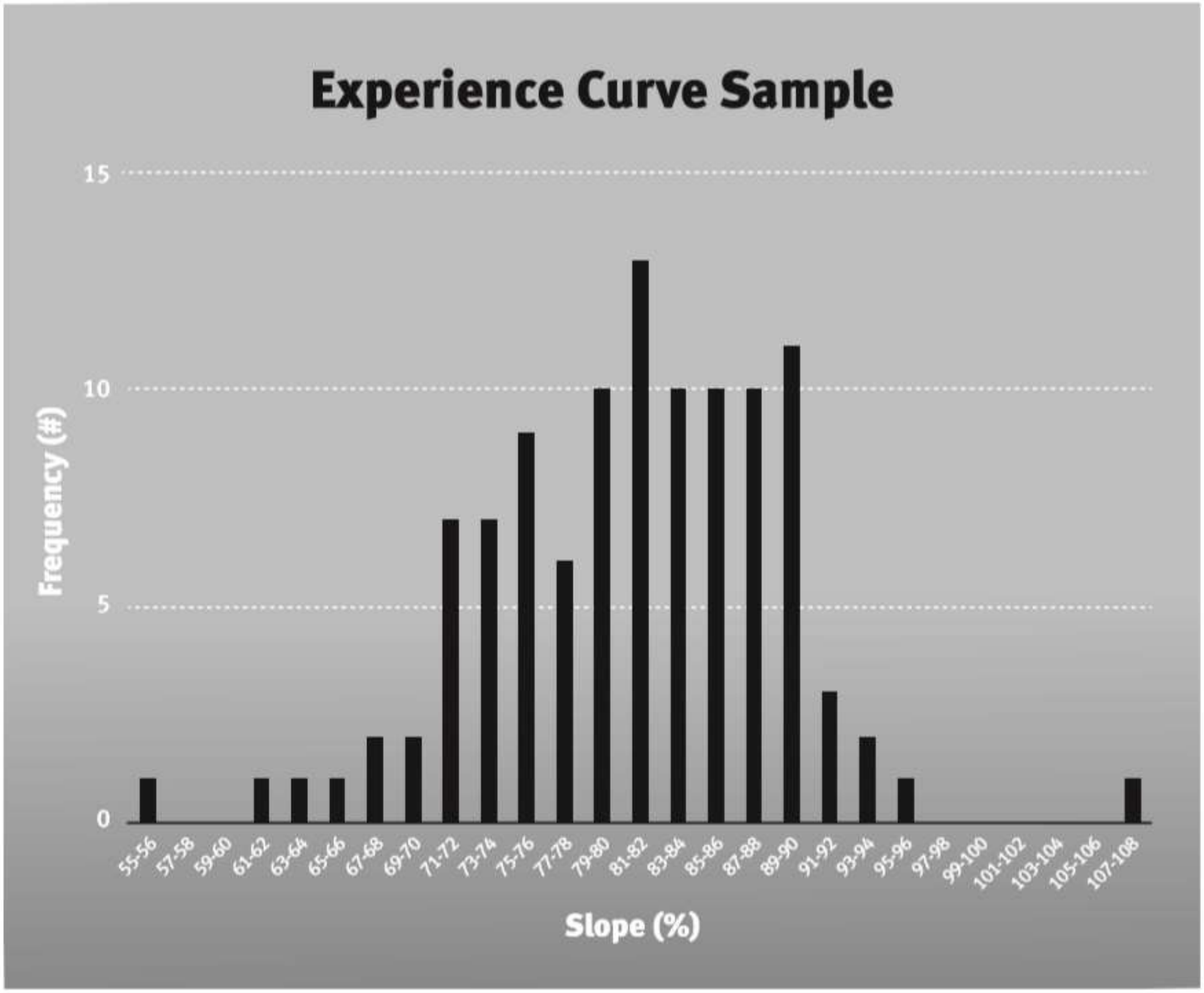
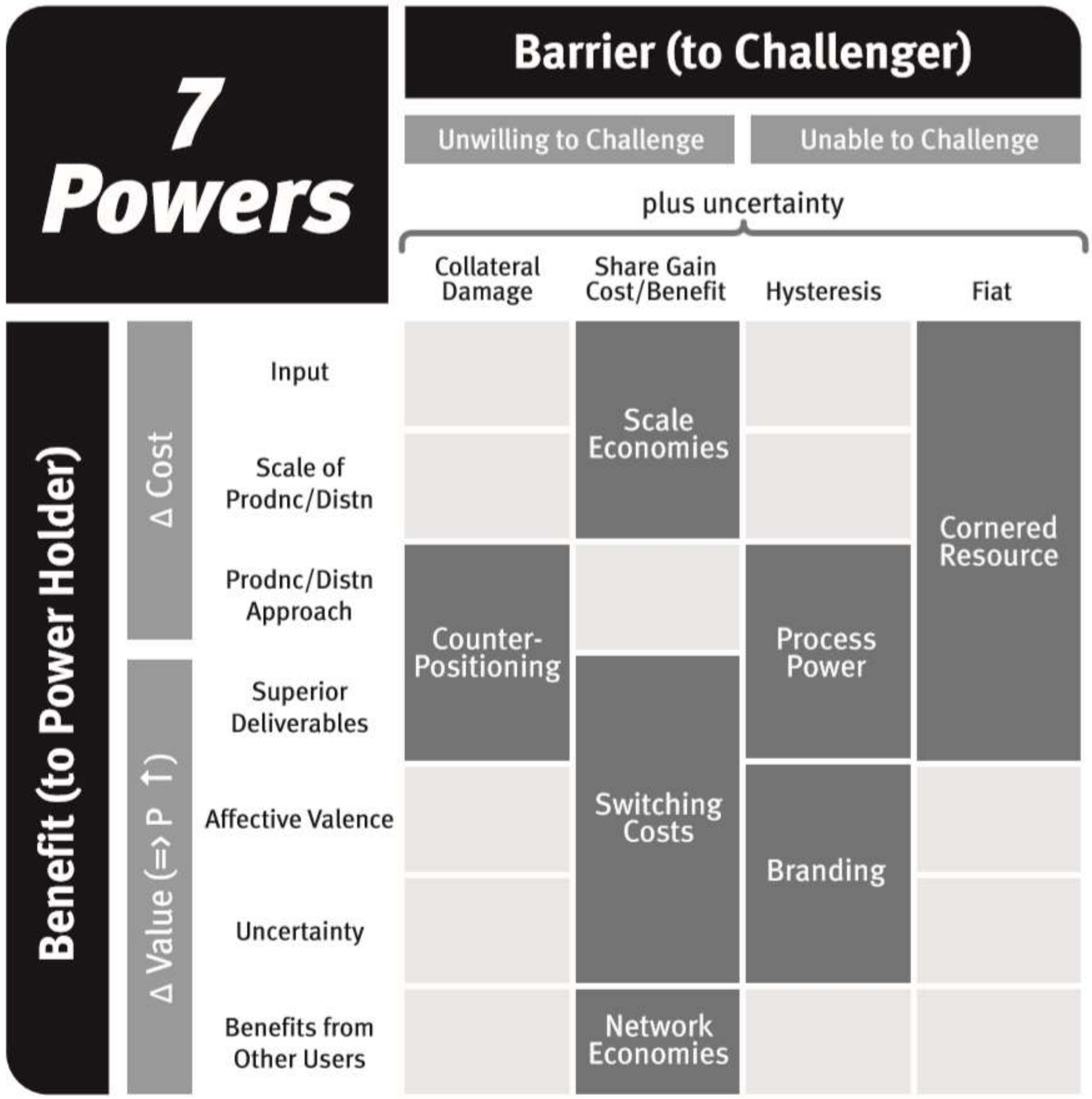


Figure 7.5: Power Intensity Determinants

	Industry Economics	***Competitive Position***
Scale Economies	Scale economy intensity	Relative scale
Network Economies	Network effect intensity	Absolute difference in installed base
Counter-Positioning	New business model superiority + collateral damage to old	Binary: entrant—new model; incumbent—old model
Switching Costs	Magnitude (intensity) of switching costs	Number of current customers
Branding	Time constant and potential magnitude of Branding effect	Duration of brand investing
Cornered Resources	Price &/or cost increment due to CR	Preferred access at a non-arbitraging price
Process Power	Time constant and potential magnitude of Process Power effect	Relative duration of Process Power advances

Figure 7.6: The 7 Powers



Appendix 7.1: Process Power Surplus Leader Margin

To calibrate the intensity of Power, I ask the question "What governs the leader's profitability of the company with Power (S) when prices are such that the company with no Power (W) makes no profit at all?"

In the case of Process Power, I assume all costs are marginal, so the zero-challenger profit price equals marginal costs. I focus on the case where the leader costs are lower due to its Process Power (alternately, it might charge a higher price due to its Process Power, or both). As a consequence:

$$\text{Profit} \equiv \pi = [P - c]\, Q$$

with $P \equiv$ price

$c \equiv$ marginal cost per unit

$Q \equiv$ units produced per time period

To determine SLM, assess:

What are S's margins if P is set $\ni {}_{W}\pi = 0$?

$${}^{W}\pi = 0 \Rightarrow \qquad 0 = (P - {}^{W}c)\, {}^{S}Q \qquad \Rightarrow P = {}^{W}c$$

Suppose:

$D(t) \equiv$ W's cost multiple at time t

$Z \equiv$ maximum potential cost multiple

$F \equiv$ cycle time compression factor

${}^{W}c$ are a multiple of ${}^{S}c$, so ${}^{W}c = D(t)^{*}\, {}^{S}c$

$$\therefore \qquad {}^{S}\pi = [P - {}^{S}c]\, {}^{S}Q = [D(t)^{*}\, {}^{S}c - {}^{S}c]\, {}^{S}Q$$

$${}^{S}\pi = (D(t)-1)\, {}^{S}c\, {}^{S}Q$$

$${}^{S}\text{Margin} = (D(t)-1)\, {}^{S}c\, {}^{S}Q / (D(t)\, {}^{S}c\, {}^{S}Q) = 1 - 1/D(t)$$

or $\boxed{\textbf{SLM} = \mathbf{1 - 1/D(t)}}$ with $D(t) = Z/(1 + (Z-1)e^{-Ft})$

Industry economics define the function D(), which determines the potential magnitude and sustainability of leverage. D(t) is an increasing function of t, reflecting that Process Power requires action over time to be increased. The logistic function was chosen to reflect the reinforcing aspect of Process Power investment, while allowing diminishing marginal returns over time. The particular form specified above for D(t) ensures that D(t)=1 at t=0 by adjusting the location parameter as a function of F and Z. When F is larger, the logistic curve is steeper and the Process Power cycle time is shorter. When F is smaller, the logistic curve is shallower and the Process Power cycle time is longer.

Time t represents the competitive position that S has relative to W in developing Process Power, because it determines the ability of a competitor starting at time t = 0 to catch up to the strong firm at time t=t. Z and F (i.e., D[t]) determine industry position. As can be seen in Figure 7.6, the longer the life cycle, the farther behind the weak competitor, and so the harder it is to catch up to the strong firm. The sustainability of the process depends on the shape of the D(t) function relative to D(0) for all t in the case of the weakest competitor with no Process Power, and an alternative D'(t) function (i.e., an alternative F') and alternative t for another competitor.

Figure 7.6: Process Power as a Function of Time

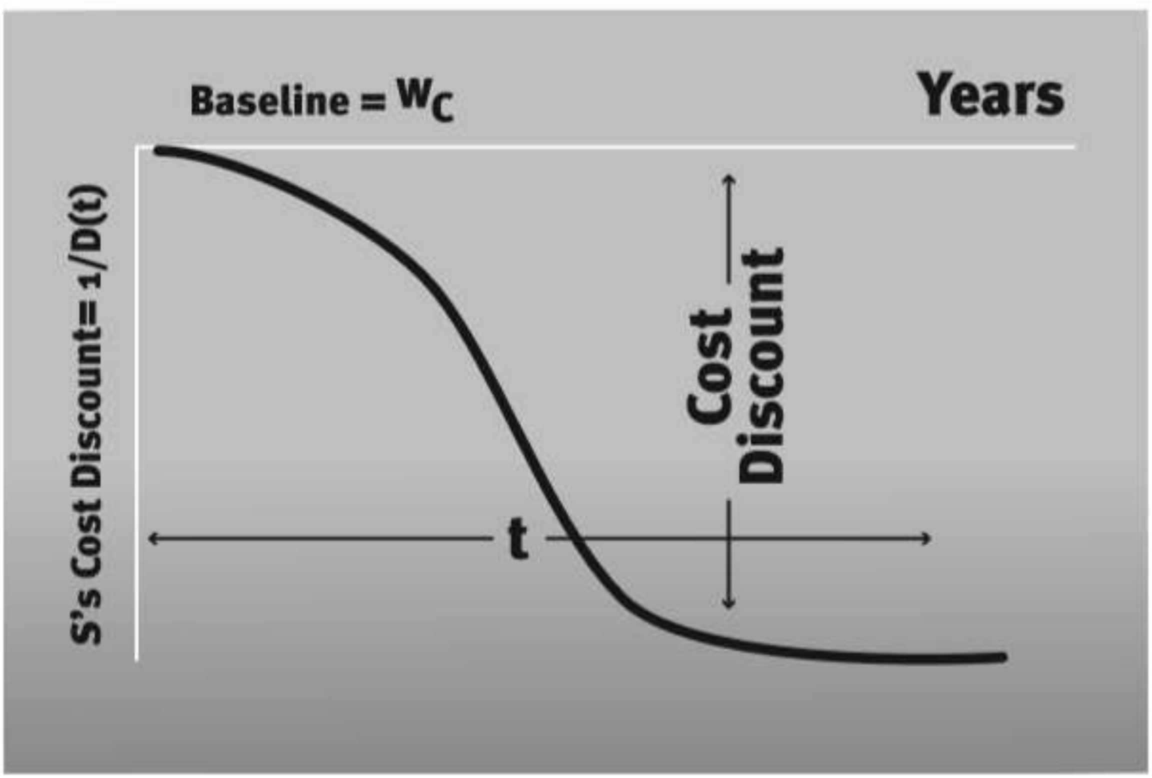


Figure 8.1: Accelerating Net Subscriber Additions (in millions)

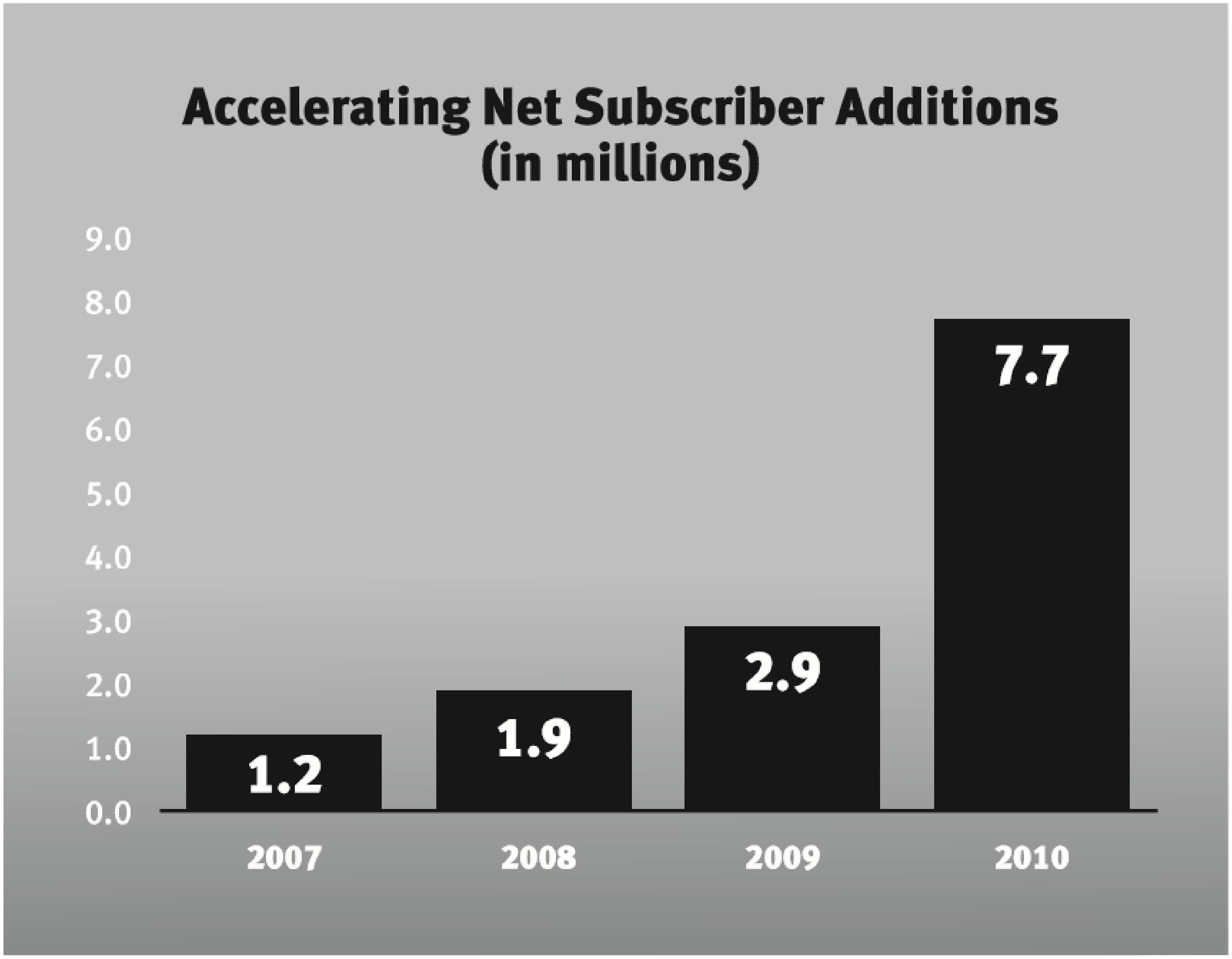


Figure 8.2: Netflix Originals[84]

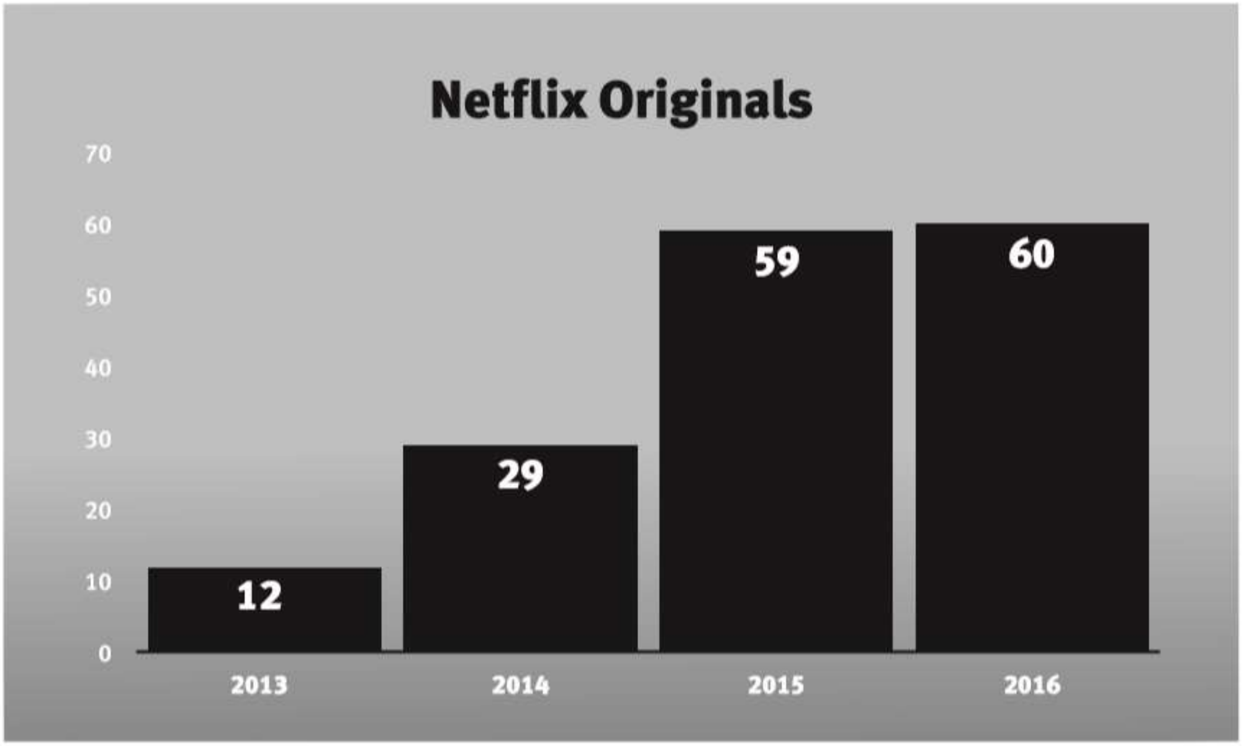


Figure 8.3: Netflix Stock Price[85]

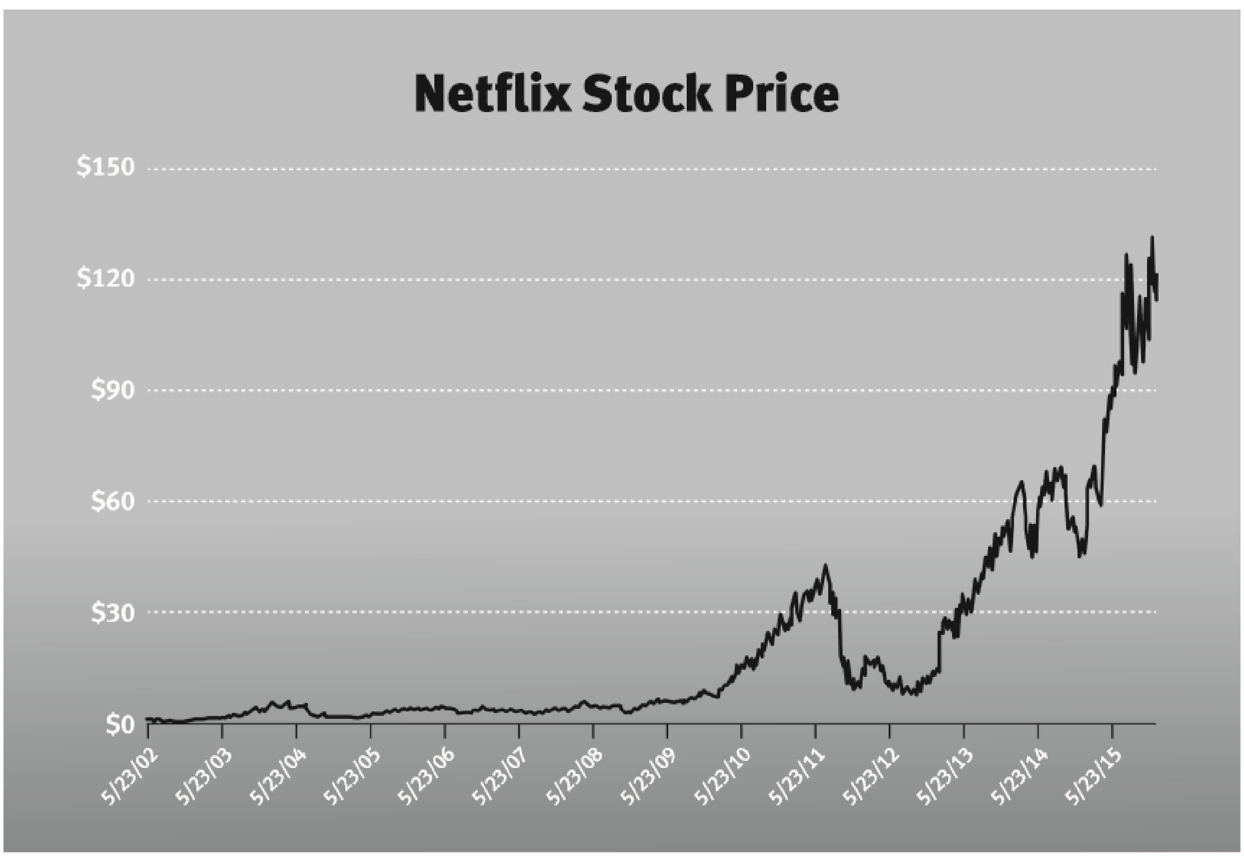


Figure 8.4: The Dynamics of Power–1

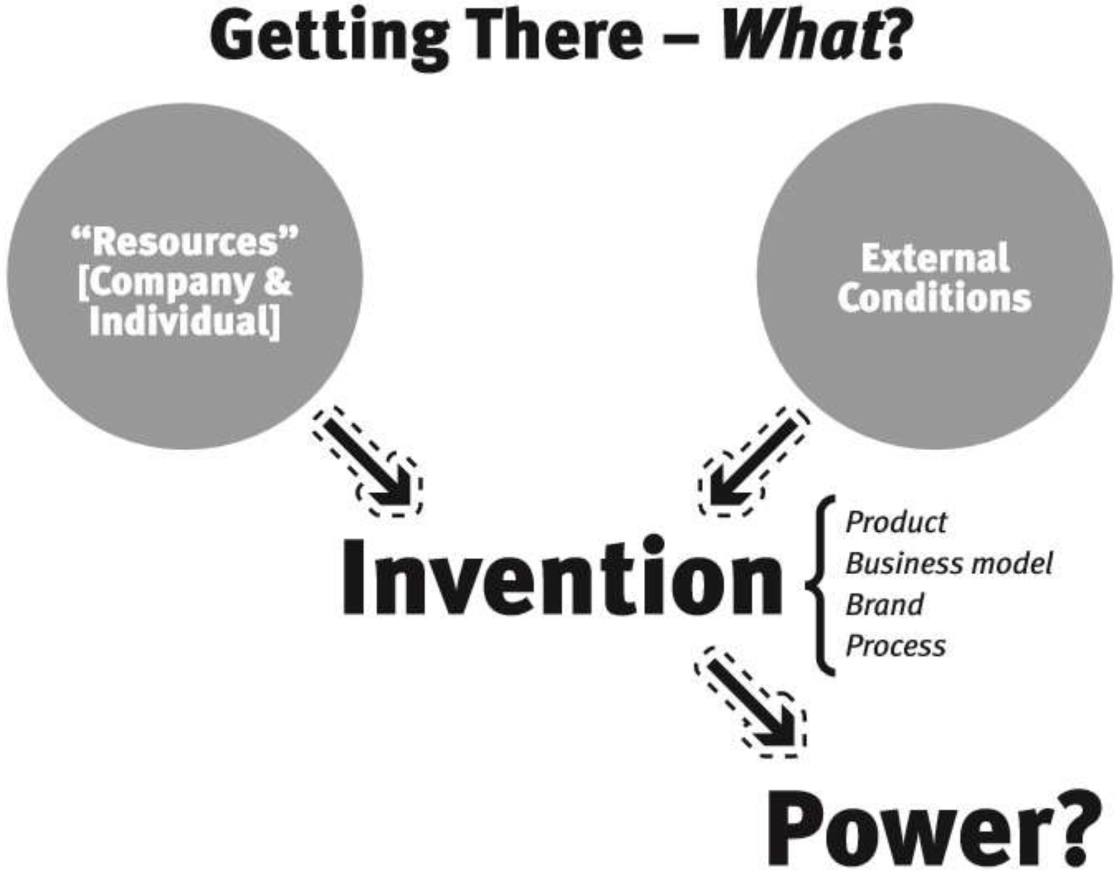


Figure 8.5: The Dynamics of Power–2

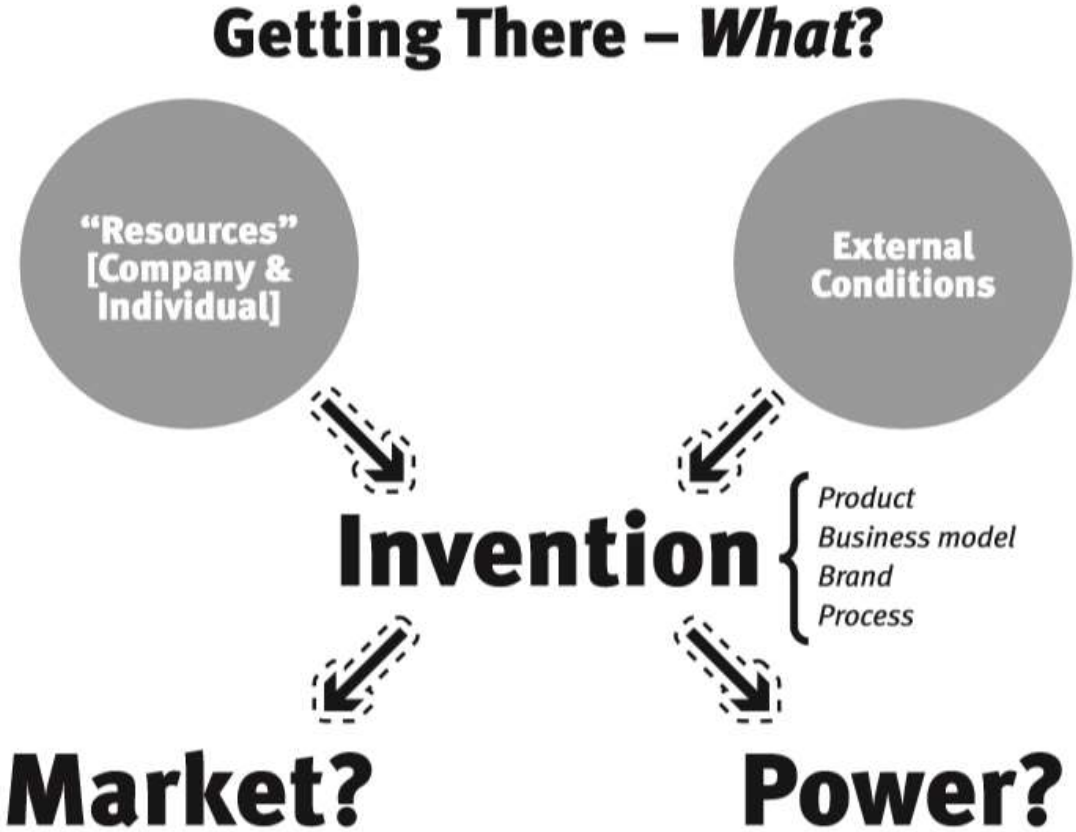


Figure 8.6: Compelling Value

Figure 8.7: Capabilities-Led Invention

Figure 8.8: Customer-Led Invention

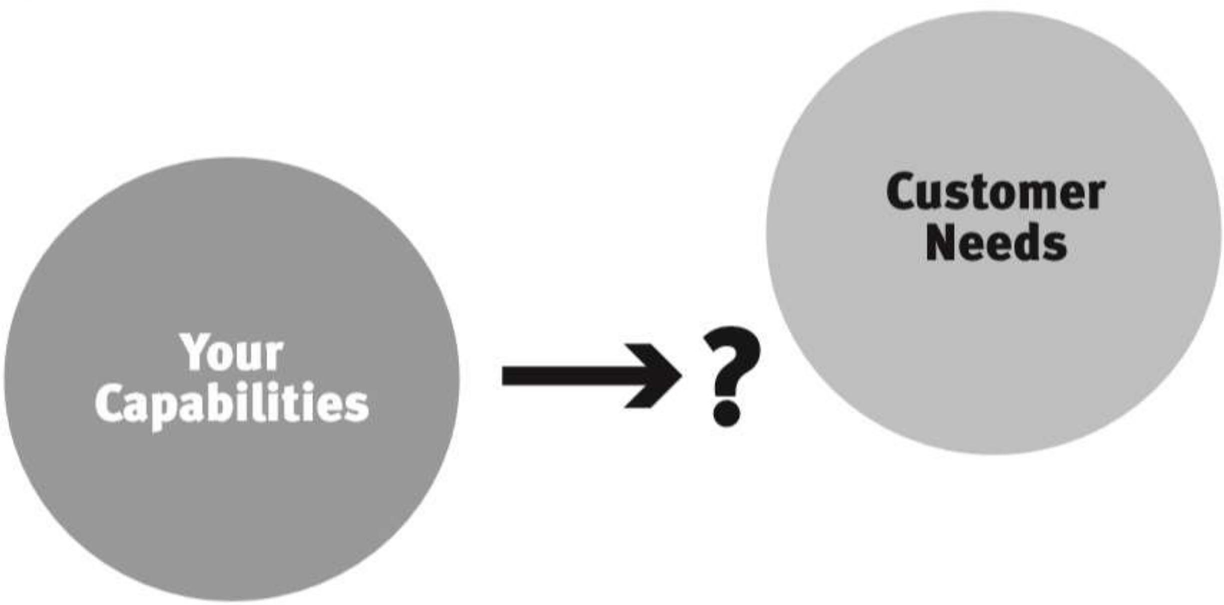
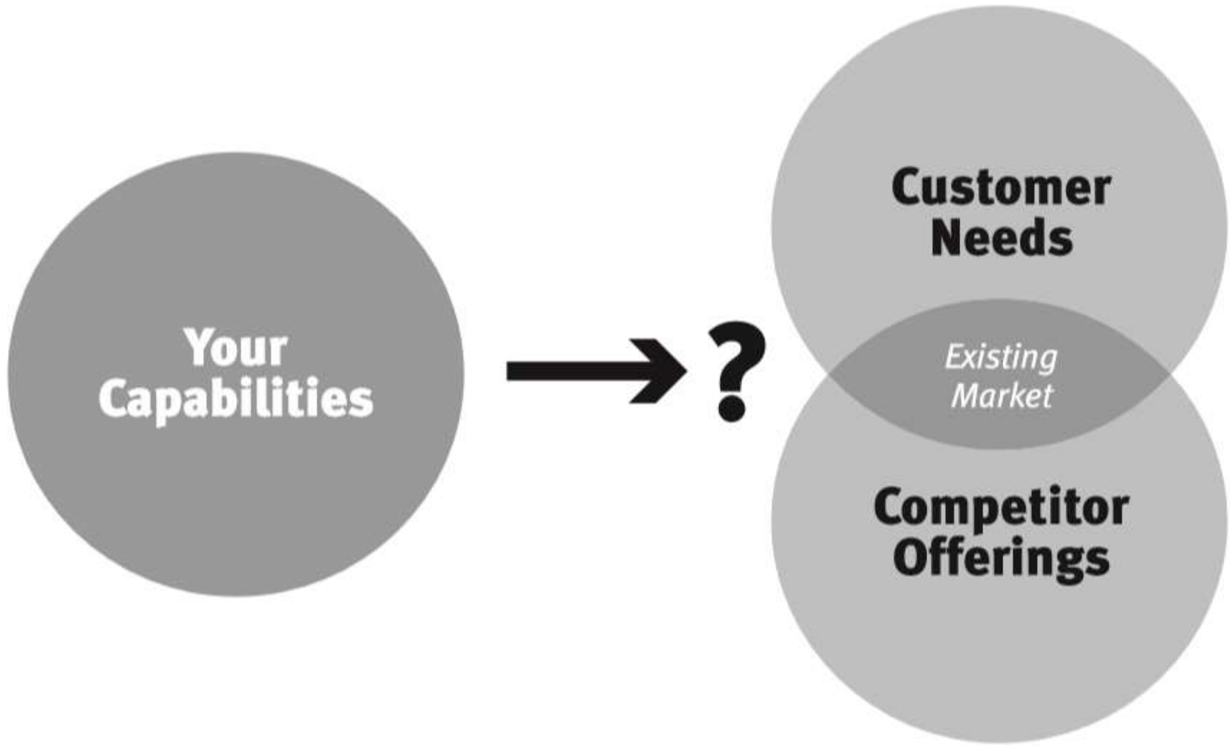


Figure 8.9: Competitor-Led Invention



Appendix 8.1: Equity Investing and the 7 Powers as a Strategy Compass

In addition to my career as a strategy advisor, I have also been an active equity investor for decades, utilizing the understanding of business value I have gained from Power Dynamics, the full Strategy toolkit I have developed which includes the 7 Powers. The Power Dynamics Toolkit is described in Appendix 9.1. My investment results over this extended period have some relevance to the themes of this book. I have made investments predicated on the differential acuity of the 7 Powers framework in correctly characterizing the prospects for Power in high flux situations. But assessing Power prospects ex ante in high flux situations is also what drives the business person's need for a strategy compass. Let me give you some details.

First, to summarize the strategy compass thesis:

- I have made the foundational assumption that Strategy and strategies are about only one thing: potential fundamental business value. I refer to this as the Value Axiom, and it is the bedrock of Power Dynamics and the 7 Powers. This assertion represents an intentional narrowing on my part. The last several decades have proven to me that much acuity and usefulness results from adopting the Value Axiom.

- By far the most important "value moment" for a business occurs when the bars of uncertainty are radically diminished with regards to the Fundamental Equation of Strategy, market size and Power. At that moment, the cash flow future makes a step-change in transparency.

- It is the period of invention, with all its high flux, that gives rise to this "value moment," offering the potential for traction in both market size and Power. High uncertainty persists during this interval because these transitions are typically not linear and quite difficult to forecast accurately.

- Strategy (the discipline) can only contribute in this period if it serves as a strategy compass to guide on-the-ground "inventors," increasing their likelihood of finding a path to satisfy The Mantra.
- To serve as such a cognitive guide, a Strategy framework must be simple but not simplistic. That is the objective of the 7 Powers.

So what relevance does this bear to active equity investing?

To the best of my knowledge, the 7 Powers applies to all businesses, everywhere. Furthermore, it is founded on fundamental business value, which also concerns a large class of investors. Does this mean that utilizing the 7 Powers can result in alpha[99] for investment in any company? Of course not.

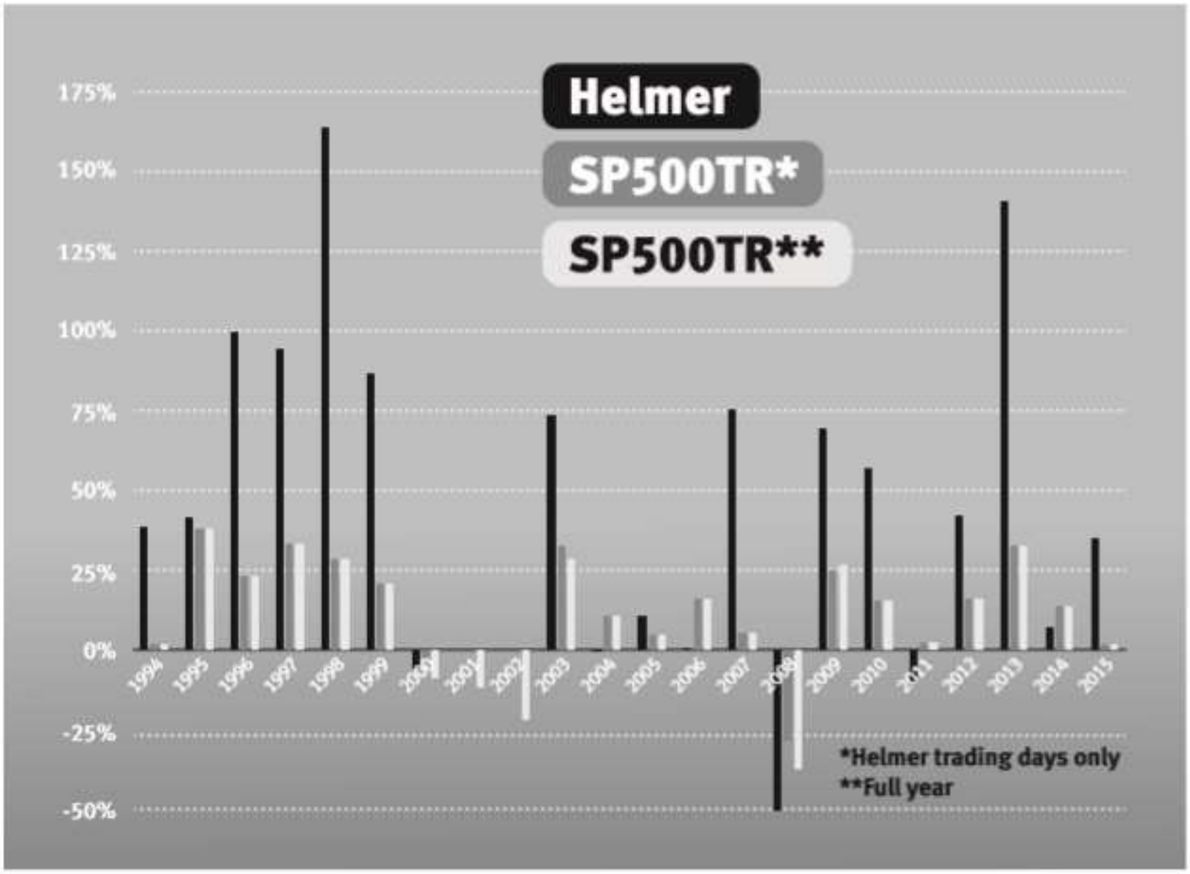
In nearly all cases, both the potential for Power and the size of the market are sufficiently evident to astute investment professionals. In particular, they can often be found in the markers of historical financials. Alpha depends on exceptions to the semi-strong form of the Efficient Market Hypothesis: you need material informational advantage. In these cases the 7 Powers offers no such advantage.

The only places one might expect alpha from applying the 7 Powers are those situations in which such transparency is not the case—opacity in other words—and that such opacity is penetrable by the 7 Powers.

A primary driver of opacity is high flux: if a business is in a fast-changing environment, then the information facing investment pros tends to have much higher uncertainty bars regarding future free cash flow. But high flux also attends the sort of conditions which orbit the "value moment." So if the 7 Powers can lead to alpha by identifying Power in these situations *ex ante*, it also promises to be useful in doing the same for those inventors on the ground trying to find a path to satisfy The Mantra.

So how have I done following this approach? Have I been able to deliver alpha? My active investing records go back twenty-two years, but I will be brief in answering those questions. I have daily portfolio returns data for all the 4664 trading days I was in the market, dating from the beginning of 1994 through 2015.[100] My year-by-year annual gross returns are shown in the chart below.

Figure 8.10: Annual Helmer Active Equity Gross Returns (1994–2015)

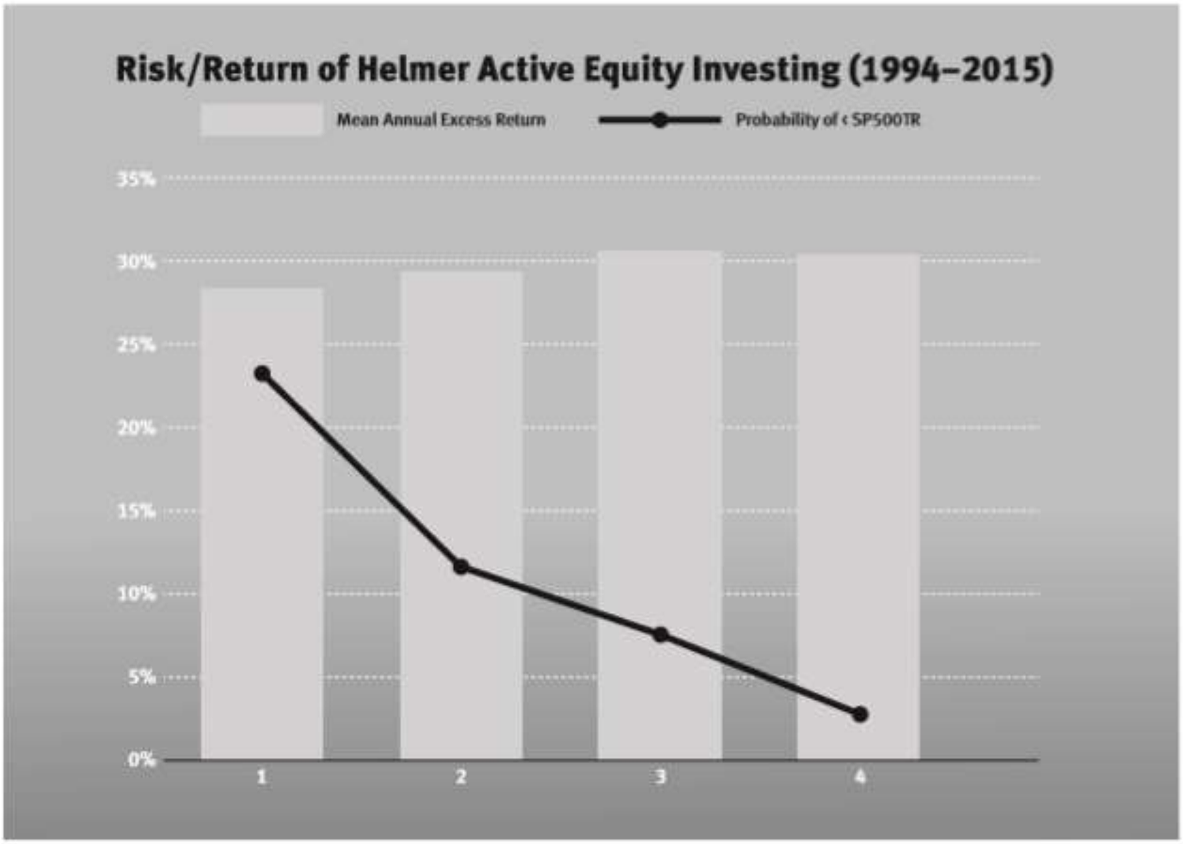


So for these 22 years, I was fully in the market for seventeen years, partially for three years and fully out of the market for two years. For the twenty invested years (seventeen full years and three partial years), my gross returns exceeded those of the market for fourteen years and underperformed the market for six. Over these trading days in which I actively invested, I achieved an average annual rate of return of 41.5% per year versus 14.9% per year for the S&P 500 TR. So on average I outperformed the S&P 500 TR by 26.6% each year.[101]

However, my highly concentrated approach results in a different risk profile than that of the market overall. So we should also look at a risk-adjusted return.

One way to do so is to take out the effects of the market overall (beta). This yields an average annual alpha of 24.3% (9.1 basis points of alpha per trading day, on average).

Figure 8.11: Risk/Return of Helmer Active Equity Investing



Years Invested

Because of the high concentration,[102] my approach has higher volatility: 31.6% versus 15.8% for S&P 500 TR when annualized. An informative way to assess the risk/reward associated with this is to calculate an investor's prospects of "beating" the market. To do this for all 4664 trading days, I calculated the risk/return profile:

1. *Risk.* What was the probability of underperforming the market if an investor had stayed with my approach for one, two, three and four years?[103]

2. *Return.* What would have been the average annual rate of return over those one, two, three and four year periods?

Figure 8.11 above displays those calculations. This chart would indicate that, historically, if you had invested based on this approach and continued with that investment for 4 years, you would have had about a 3% probability of not outperforming the market (this was your "risk"), and the average return for a four-year hold outperformed the market by about 30% per year (this was your "return").

So based on both these measures, utilizing the 7 Powers has historically delivered unusually attractive returns over an extended period. I know of no other Strategy framework with this outcome. Thus the 7 Powers seems to have been differentially acute as a tool for identifying the potential for Power *ex ante* in high flux situations. This provides additional assurance in its utility as a cognitive frame for business leaders in their crucial "value moments," which are also inevitably high flux.

Figure 9.1: Intel Stock Price vs. S&P 500 TR (Index Value: 3/17/1980 = 1.00)[107]

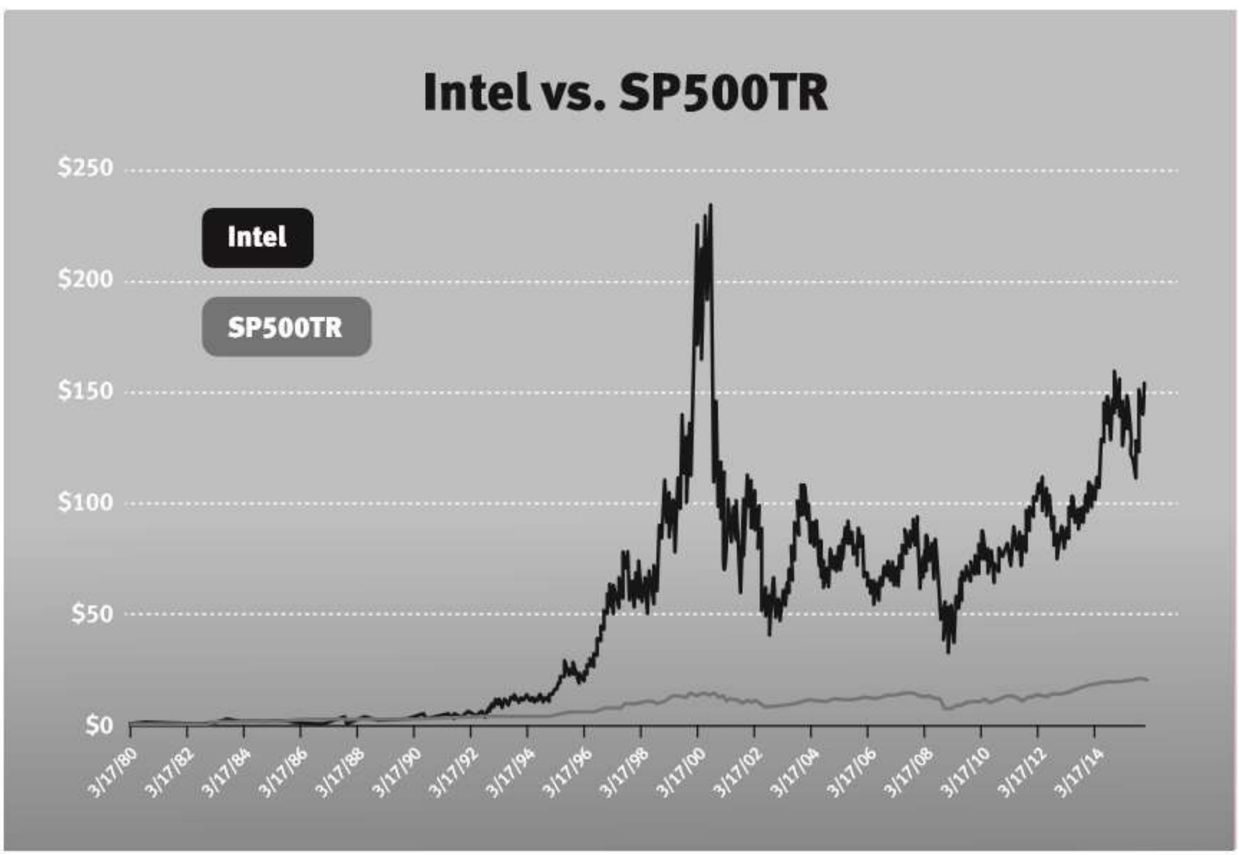


Figure 9.2: Annual Growth of Microcomputer Shipments (units)[110]

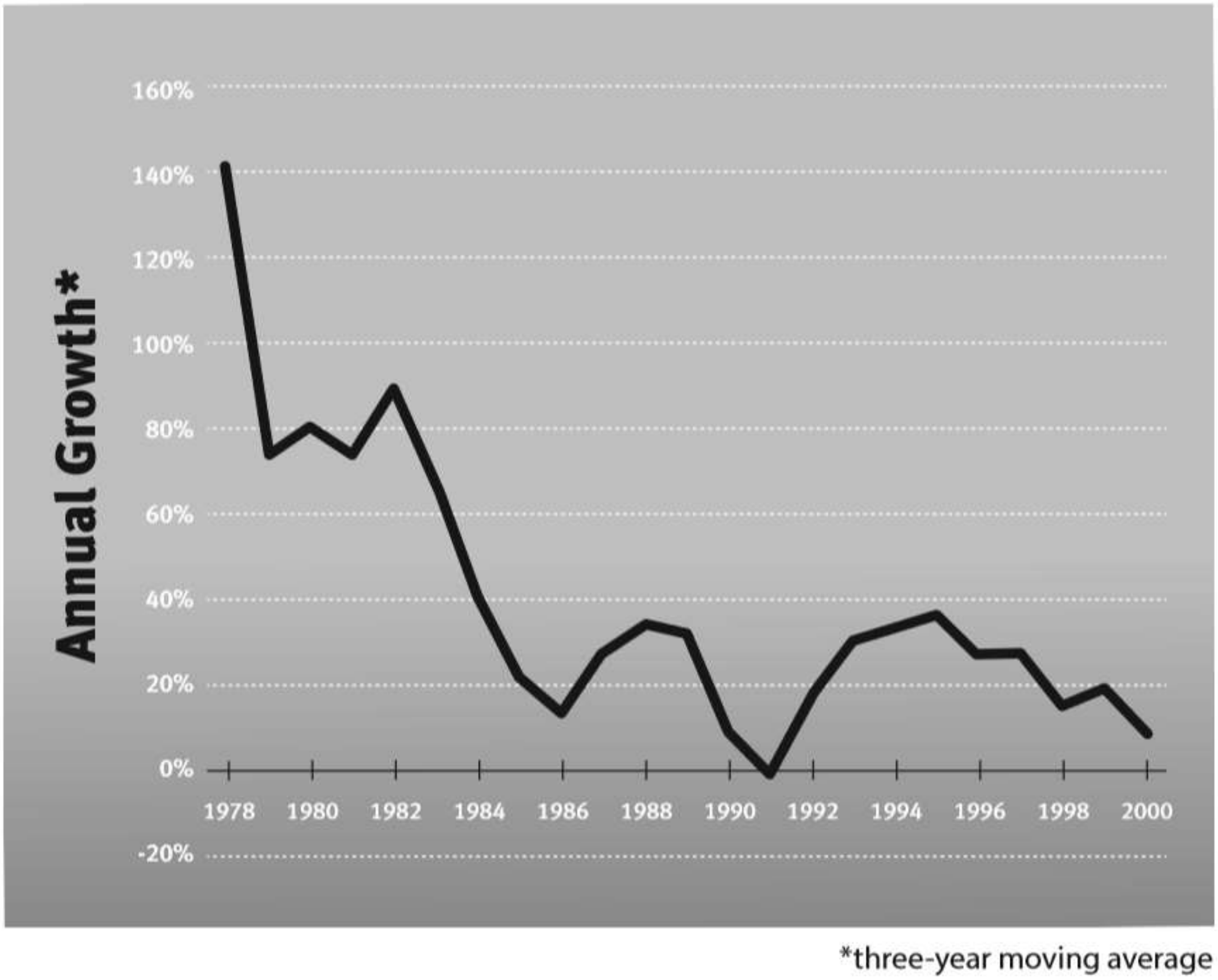


Figure 9.3

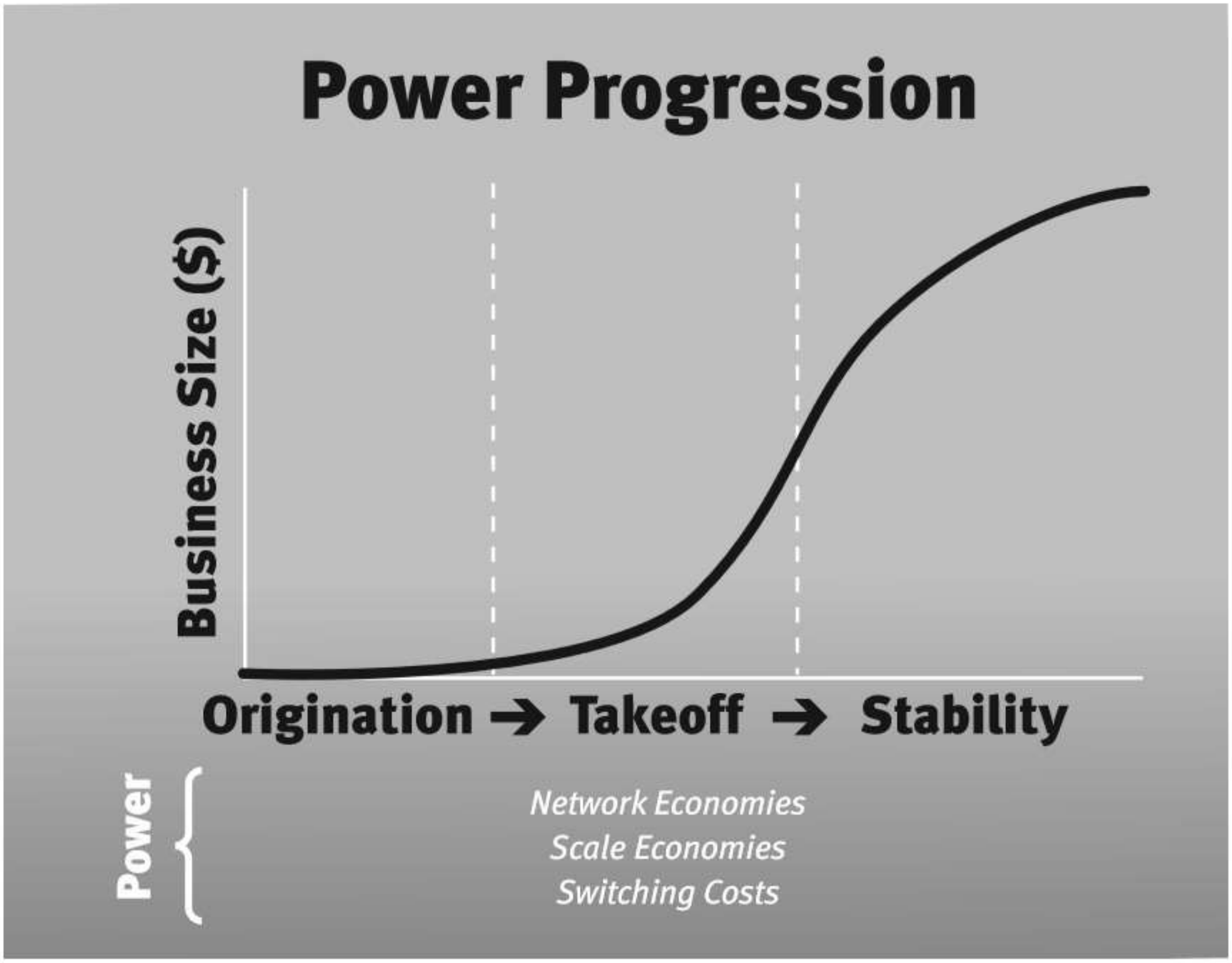
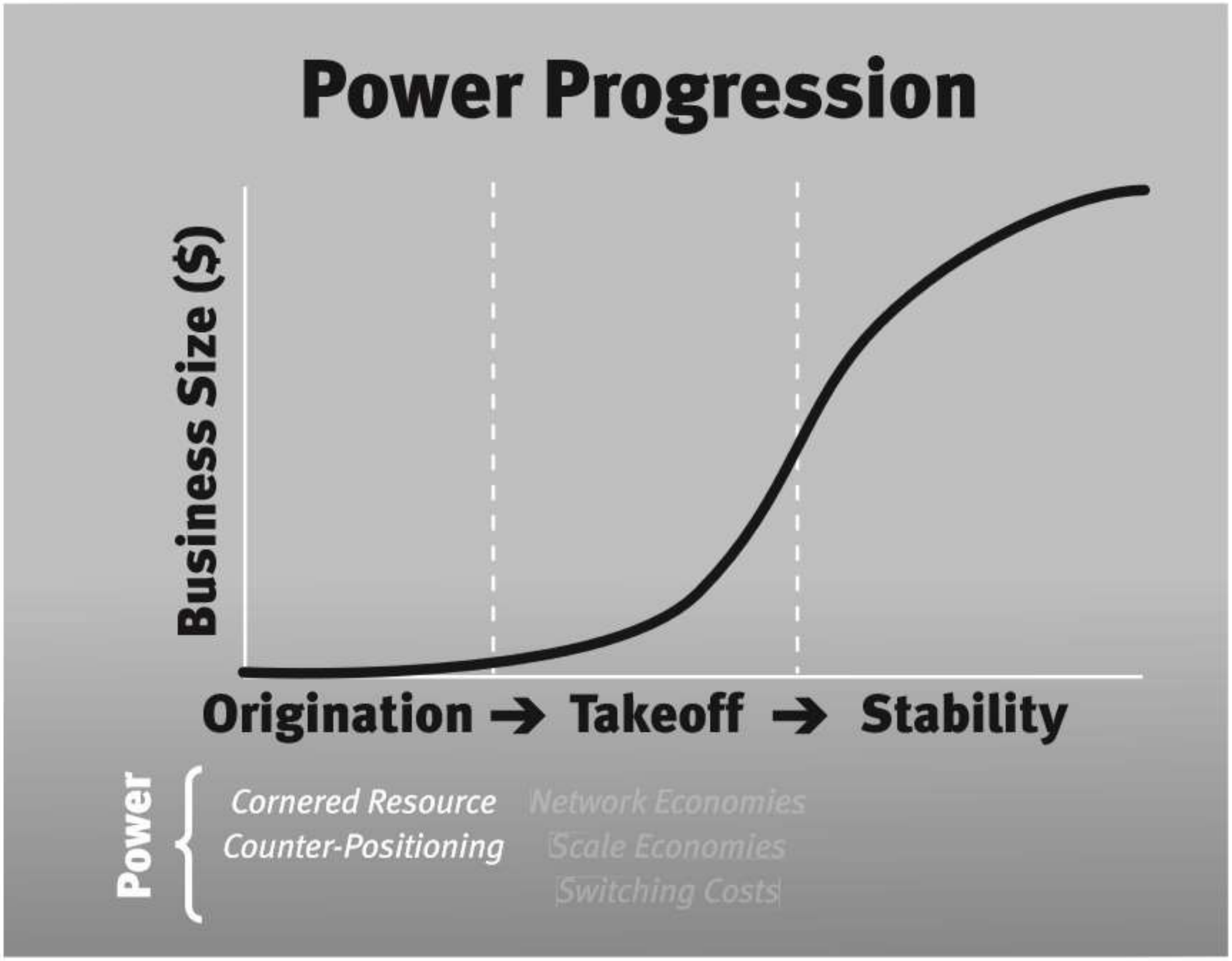

Power Progression
Business Size ($)
Origination ➔ Takeoff ➔ Stability
Power
Network Economies
Scale Economies
Switching Costs

Figure 9.4



Power Progression
Business Size ($)
Origination ➔ Takeoff ➔ Stability
Power
Cornered Resource
Counter-Positioning
Network Economies
Scale Economies
Switching Costs

Figure 9.5

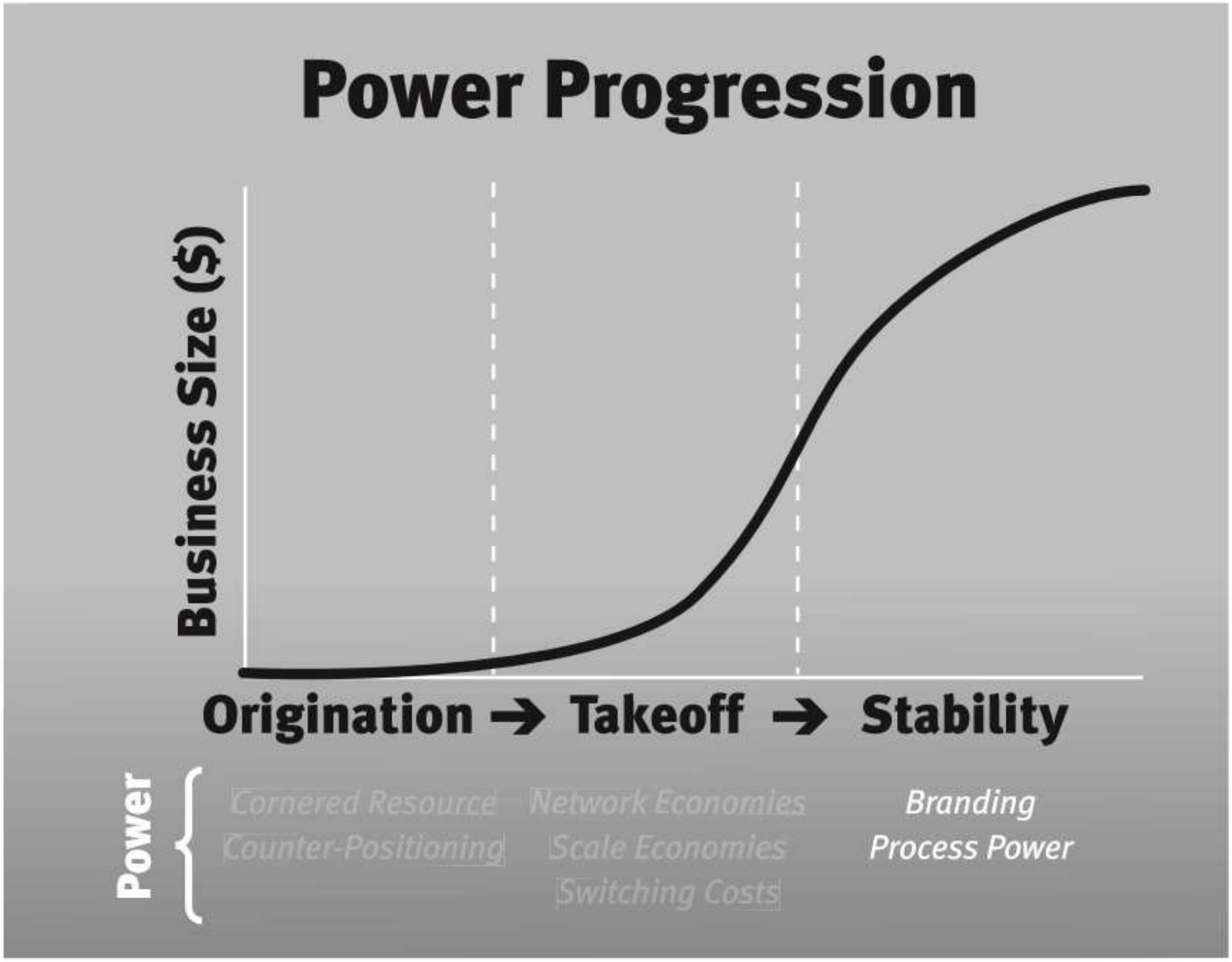

Power Progression
Business Size ($)
Origination ➔ Takeoff ➔ Stability
Power
Cornered Resource
Counter-Positioning
Network Economies
Scale Economies
Switching Costs
Branding
Process Power

Figure 9.6: Time-Delineated 7 Powers

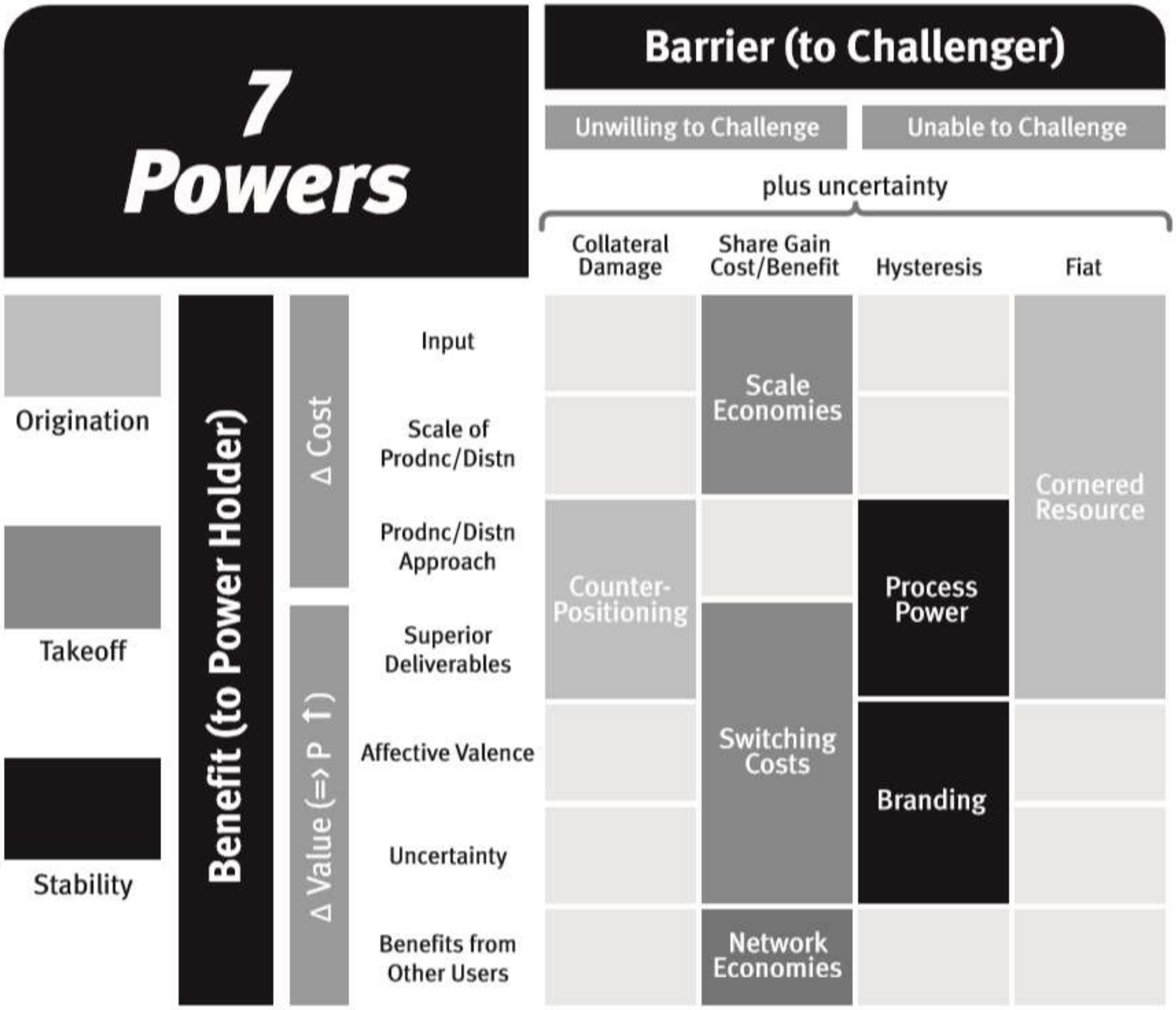
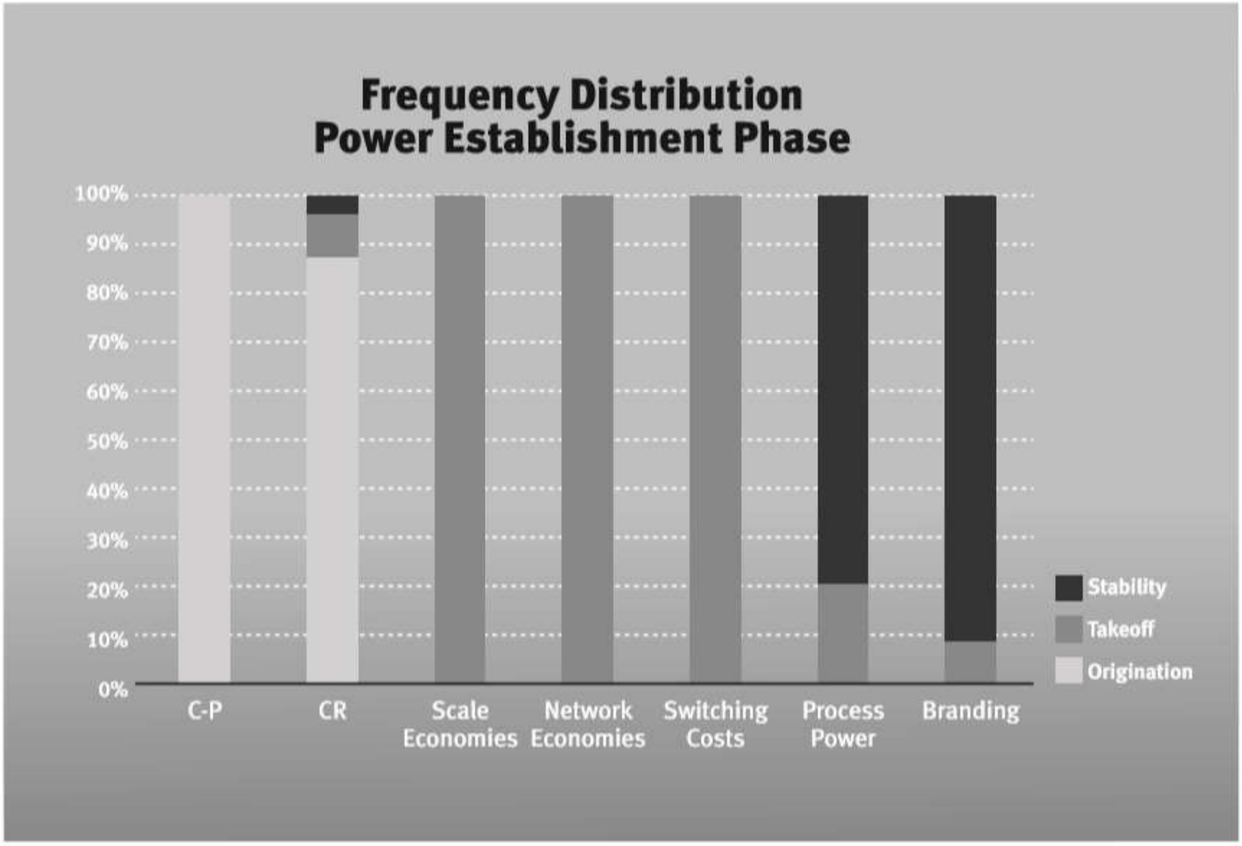


Figure 9.7


Frequency Distribution
Power Establishment Phase
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
C-P
CR
Scale Economies
Network Economies
Switching Costs
Process Power
Branding
Stability
Takeoff
Origination

Appendix 9.1: The Power Dynamics Toolkit

The body of Strategy intellectual capital I have developed is called Power Dynamics. The 7 Powers is its central unifying framework. Overall Power Dynamics is built on and tied tightly to seven perspectives.

1. **The Value Axiom.** Strategy has one and only one objective: maximizing potential fundamental business value.

Commentary. This is an assumption not a proof. My experience is that this narrowing of the scope of Strategy and strategy has a profoundly positive impact on the usefulness of the discipline. Note that this is fundamental not speculative value. Further it is about potential value. Realizing that value requires operational excellence.

2. **The 3 S's.** Power, the potential to realize persistent differential returns, is the key to value creation. Power is created if a business attribute is simultaneously:
   - Superior—improves free cash flow
   - Significant—the cash flow improvement must be material
   - Sustainable—the improvement must be largely immune to competitive arbitrage

Commentary. In this book I have focused on Benefit + Barrier which has a one-to-one mapping to the 3 S's (Superior + Significant = Benefit and Sustainable = Barrier). However, in the field, the tripartite 3 S test of Power proves useful additionally because, since it calls out "Significant" separately, it makes materiality explicit. For example, businesses often tout network effects but, when looked at carefully, they are not material and therefore do not qualify as Power.

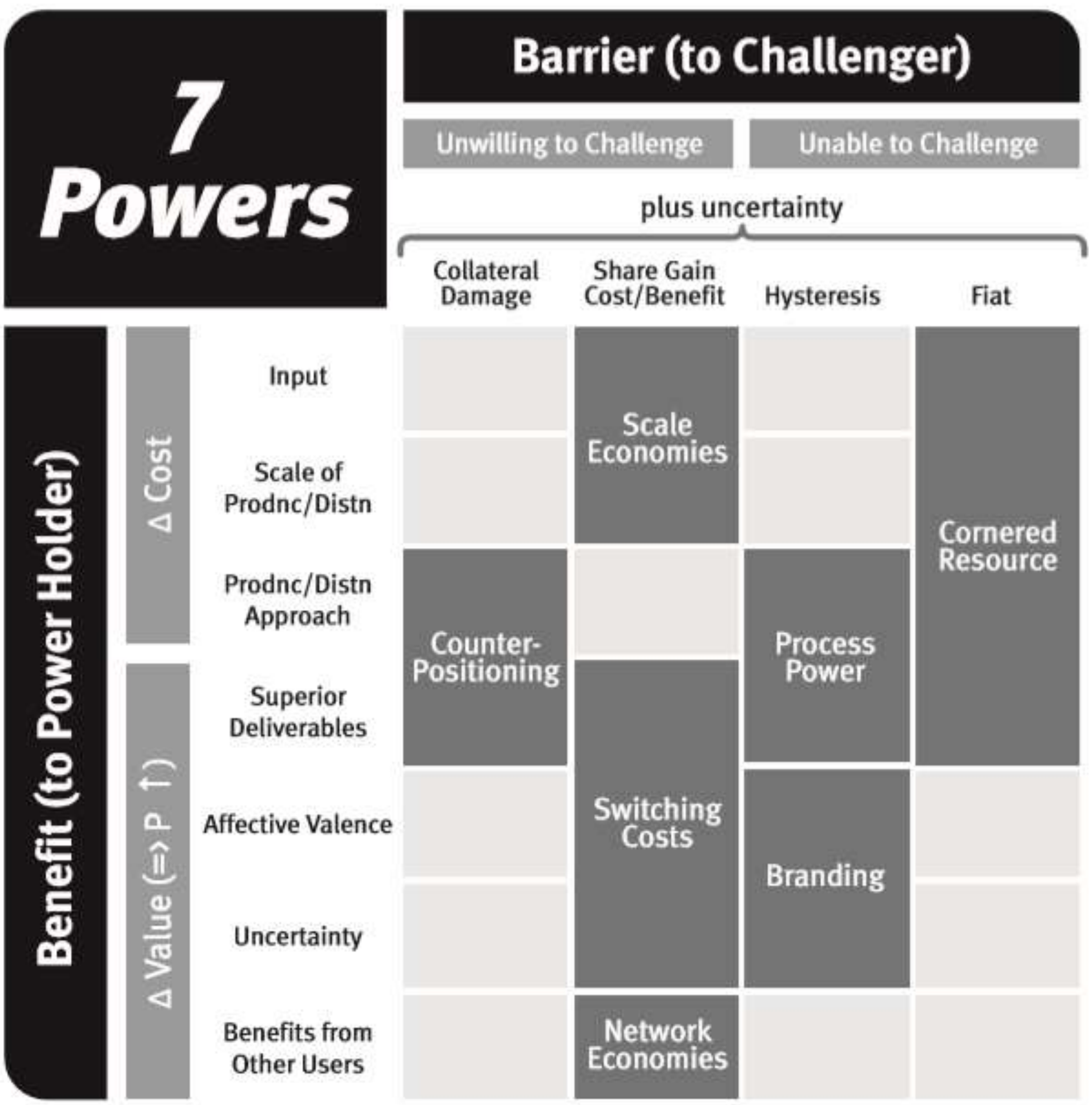
3. **The Fundamental Equation of Strategy. Value = $M_0\, g\, \bar{s}\, \overline{m}$**

Commentary. The interpretation of this is that Value = Market Size * Power. M_0 is the current market size, g is a discounted growth factor for the market, $\bar{s}$ is long-term average market share and $\overline{m}$ is long-term average differential margins (the profit margin above that needed to return the cost of capital). I have found that the explicit tying of Strategy concepts to the exact determinants of the net present value of free cash flow puts to rest a lot of fuzzy thinking about the relationship between Strategy and value. It has also helped me as an active equity investor. It i important that $\bar{s}$ and $\overline{m}$ are long-term equilibrium values. Short-term movement in these have little impact on fundamental value.

4. **The Mantra.** A route to continuing Power in significant markets.

Commentary. If you only take one phrase away from reading this book, I hope i is this one. It is a complete statement of the elements of a strategy. It maps directl to the Fundamental Equation of Strategy and is inclusive of Dynamics. The wor "continuing" is included, even though Power implies sustainability, to encourag ongoing layering on of different sources of Power as a business progresses.

5. **The 7 Powers.**

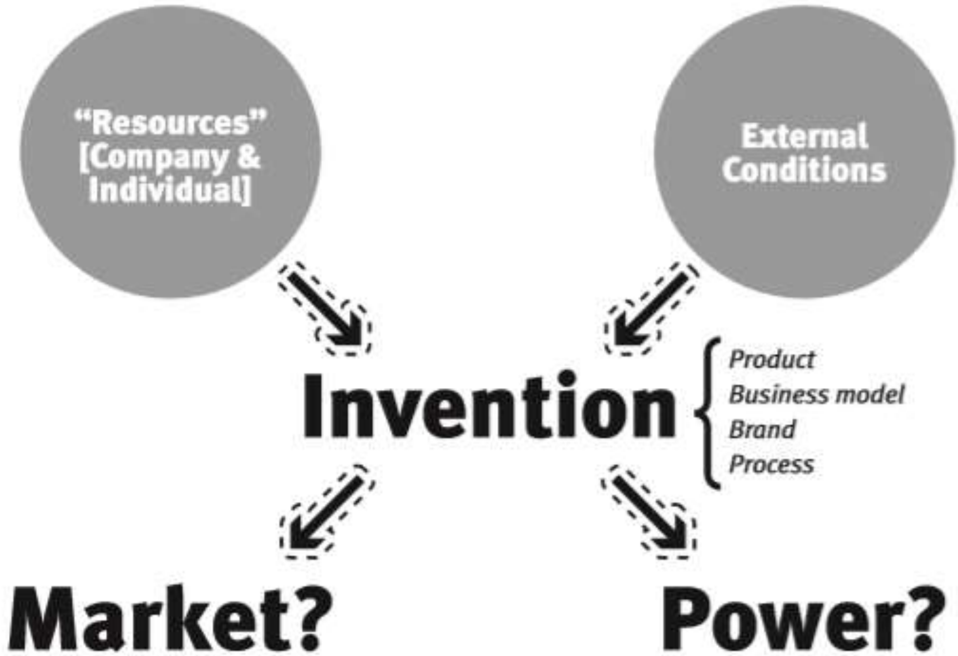


Commentary. To the best of my knowledge the seven Power types positioned on this chart are the only strategies available to a company. If you do not have at least one of these for each competitor (current and potential, direct and functional), you cannot satisfy The Mantra and hence are lacking a viable strategy. In the 200+ strategy cases I have led over my career, these seven were sufficient. This is also true of all the cases studied by my students, probably another 200 or so.

Aside from being exhaustive, two additional characteristics of the 7 Powers enhance its usefulness:

a. *Small set.* The key strategic questions for you are: (1) "What Power types do I now have?" and (2) "What Power types do I need to worry about establishing now?" The 7 Powers informs you that there are only seven possibilities for (1) and usually you can quickly rule out several. The Power Progression informs you that at any given growth stage the maximum number of new Powers that you might explore is 3. This focusing is very valuable. If you cannot see a route to one of these 7, your strategy problem is not yet solved.
b. *Observable ex ante.* The potential for these Power types is usually evident long before detailed forecasting is possible. What I have found working with early-stage companies in Silicon Valley and working with mature companies considering new directions is that it is possible to have meaningful conversations about the potential for Power at quite an early stage.[113] My investment results are also indicative of this *ex ante* transparency.

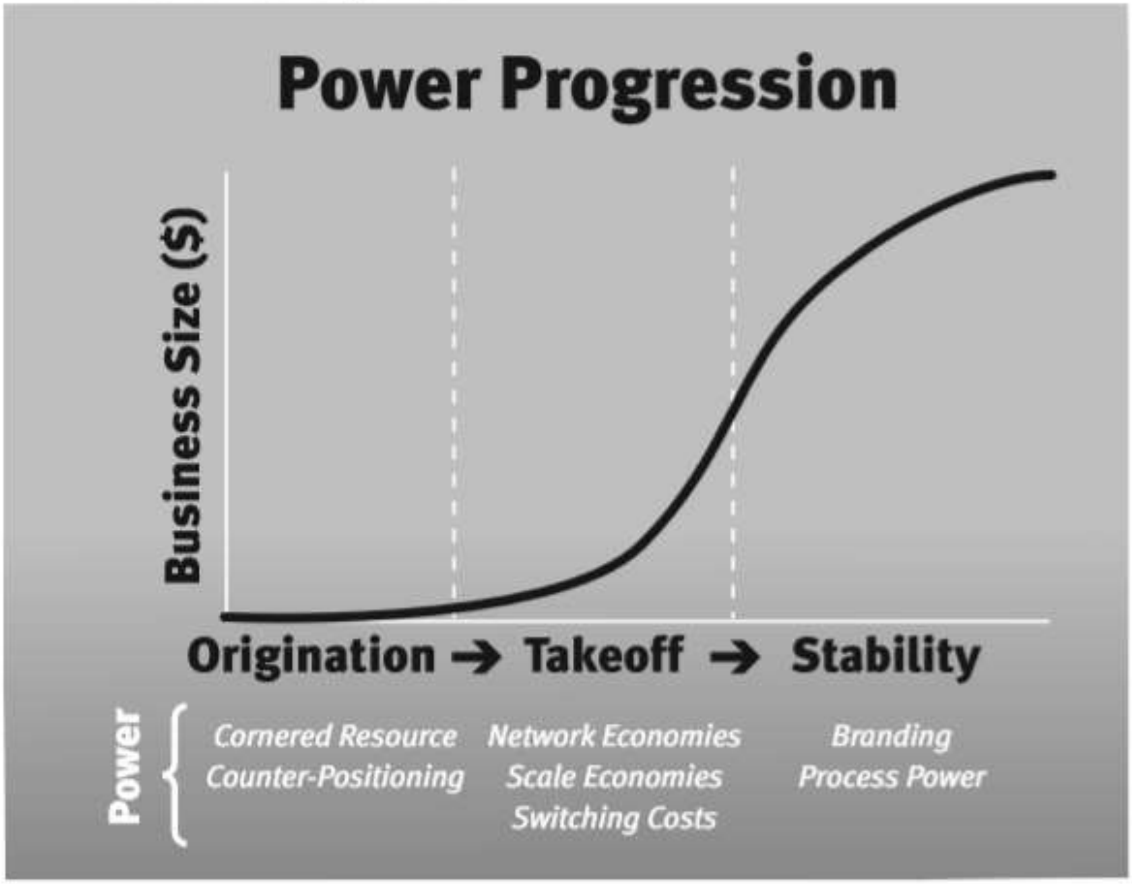
6. **"Me Too" Won't Do.** The first cause of a strategy is invention.



Commentary. Tectonic changes in value take place when Power is first established with an acceptable level of certainty. Looking at the seven Power types we can see that this always involves an invention, whether that be an invention of product, business model, process or brand. Eventually such inventions lead to a Benefit as expressed in a product attribute, be that features, price or reliability. The marker for sufficiency of such a Benefit is usually "compelling value," eliciting a "gotta have" response. There are three paths to achieving compelling value: Capabilities-led, Customer-led and Competitor-led. These each present distinctly different tactical imperatives.

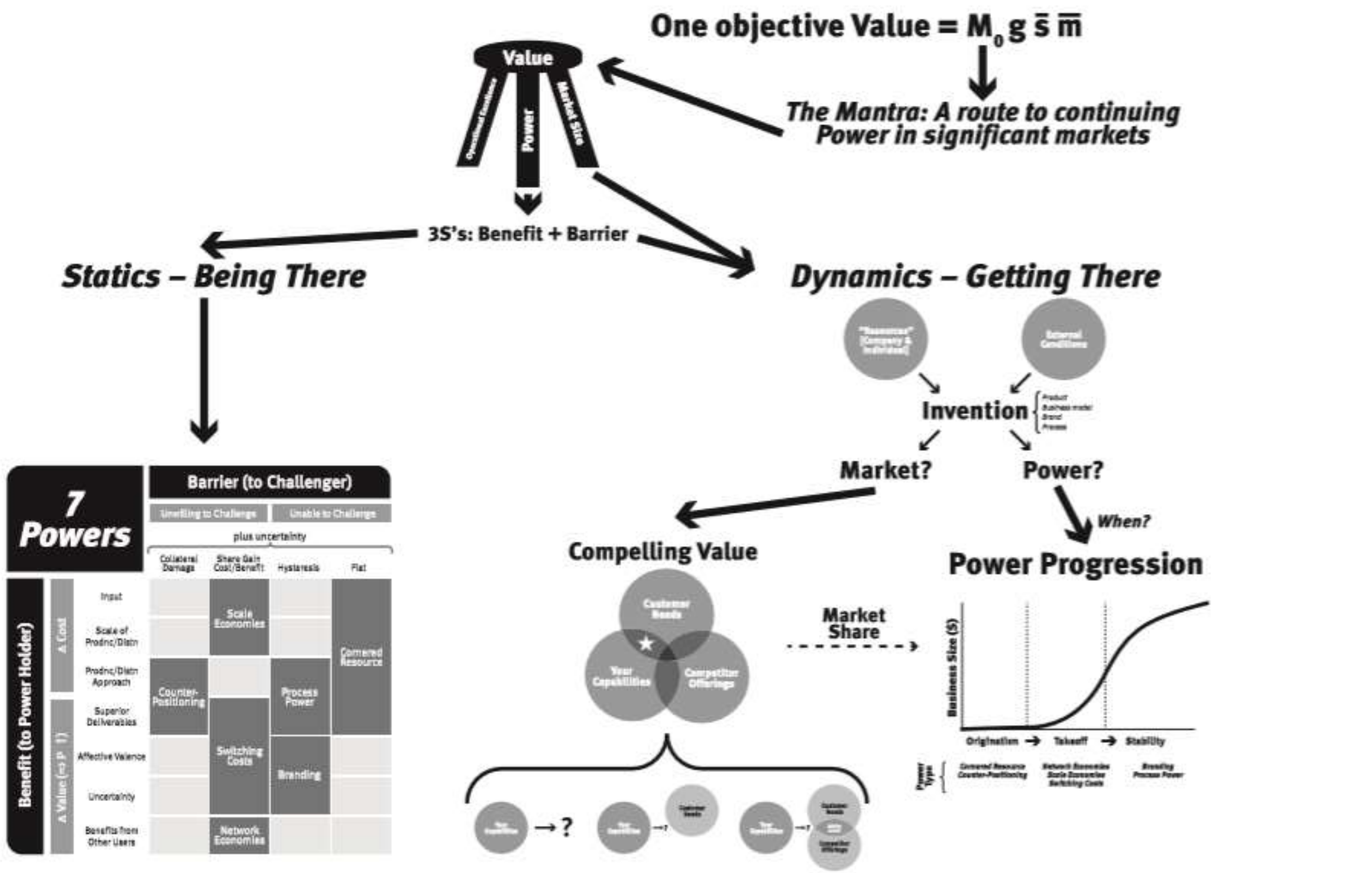
My view is that there is an important welfare implication in these relationships as well. Not only is invention the gateway to Power but also the possibility of Power (and the associated durable success) fuels invention. For example, if there were no prospects for Power then I doubt that Silicon Valley would have come to be. So from a static viewpoint, the search for Power may seem like a zero sum game of preventing gains flowing to consumers. But from a Dynamics viewpoint, it is the possibility of Power that is a critical motivator of invention. An invention only gains traction if customers flock to it. This take-up is of course a sure marker of increases in consumer welfare—they are voting with their feet. This Dynamics perspective is of course the one that should motivate policy makers.

7. **The Power Progression.**



Commentary. Different Power types present the opportunity for first establishing a Barrier at different times in the development of your business. Knowing when this window is open and when it shuts is valuable in recognizing and seizing the opportunity. The break between takeoff and stability is when unit growth falls below about 30%–40% per year. This is a business stage framework and should not be confused with the product stages of Introduction/Growth/Maturity/Decline of the Product Life Cycle which have dramatically different points of phase separation. Origination can include pre-product periods which are not covered in the Life Cycle model, and the Growth stage in the Life Cycle model includes takeoff and parts of stability in the Power Progression. These differences really matter in assessing the availability of Power.

Appendix 9.2: A graphical representation of the tools of Power Dynamics and their relationship.



Appendix 9.3: Power Dynamics Glossary

Term	***Description***
Strategy	Strategy (with a Capital S) is the intellectual discipline sometimes called Strategic Management. I define it: the study of the fundamental determinants of potential business value.
Power	The set of conditions needed for persistent differential returns. Power requires both a Benefit, something that materially increases cash flow, and a Barrier, conditions such that all the value to the firm of the Benefit is not arbitraged out by competition.
strategy	Strategy (with a lower case s) is the path to potential value for a strategically separate business. I define it: a route to continuing Power in significant markets.
value	The fundamental enterprise value of an activity. This is reflected *ex post* as generation of accessible returns to an owner (free cash flow). It is investors' expectation of the stream of these returns discounted over time that determines value *ex ante*.
Strategy Dynamics	The study of strategy development over time.
Strategy Statics	The study of strategic position at a single point in time.
industry	The group of businesses whose products have a high degree of substitutability.
business	A strategically separate economic activity. By strategically separate I mean that its Power position is largely orthogonal to the Power position of other activities the firm pursues.
market	The revenue attributable to all firms in an industry.

Term	Description
industry economics	The economic structure of a particular industry. For example, with fixed-cost-driven Scale Economies, this is measured by the magnitude of the fixed cost relative to the company's overall financials.
competitive position	A characterization of a company position in the metric relevant to Power. For example, with Scale Economies it is relative scale compared to the largest competitor.
Surplus Leader Margin	The profit margin a Power holder will achieve if pricing is such that a competing firm with no Power has zero profits. This is not necessarily an expected equilibrium but rather is a good marker of the leverage possessed by the Power holder. This would equal $\overline{m}$ in the Fundamental Equation of Strategy if the firm without Power experienced competitive arbitrage that resulted in its earning just its cost of capital and that the cost of capital for the firm with Power was equal to the cost of capital of the firm without Power.